

TENSION RESPONSES TO SUDDEN LENGTH CHANGE IN STIMULATED FROG MUSCLE FIBRES NEAR SLACK LENGTH

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SUMMARY

1. Apparatus for applying a step change of length to an isolated muscle fibre is described. The step was complete in about 0.2 ms.

2. Effects of tendon compliance were eliminated by using a spot-follower device and by gripping the tendons with metal clips close to the fibre ends.

3. The natural frequency of the force transducer was above 10 kHz.

4. Steps of various amplitudes and in either direction were applied to isolated muscle fibres about 6 mm long from the anterior tibial muscle of *Rana temporaria* during tetanic stimulation. Initial sarcomere length was 2.0–2.2 μm , and temperature was 0–3 °C.

5. The tension response to a step could be divided into four phases. The initial response was an apparently elastic change during the step itself (phase 1). After the step was completed there was a rapid partial recovery towards the original tension (phase 2, lasting 2–5 ms), followed by a slowing or reversal of recovery (phase 3, 10–50 ms), and finally a much slower return to the original tension (phase 4). Most of this paper is concerned with phases 1 and 2.

6. The initial tension change (phase 1) occurred synchronously with the applied length change, indicating that the fibres possess a compliance which is almost linear and almost undamped. Its stiffness is such that an instantaneous shortening of about 4 nm per half-sarcomere would bring the tension to zero from its isometric value.

7. The absence of detectable damping during phase 1 indicates that the viscosity of a stimulated fibre is substantially less than the apparent viscosity of a fibre at rest.

8. The instantaneous force–extension curve approached the length axis at a sharp angle and a negative tension appeared at the force transducer

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when a very large step was applied. These observations suggest that the structures responsible for the stiffness of the fibre remain rigid when they are not under tension.

9. During the few milliseconds after the step (phase 2) the tension recovered part of the way toward the level which existed before the step. In shortening steps the time course of this recovery was adequately fitted by the sum of four exponential terms, and was similar in steps of different amplitude but with a time scale shorter the larger the step. In stretches the slow components were relatively larger than in releases.

10. The tension level, T_2 , approached during phase 2 depended only on the total amplitude of the step and not on the time course of the length change, provided it was complete in 1–2 ms. The extreme tension reached during a step could thus vary widely without detectable change in T_2 .

11. With stretches and releases of up to about 3 nm per half-sarcomere this early recovery was almost complete, so that the curve of T_2 against step amplitude was nearly horizontal. With larger releases the line curved downwards, reaching zero in a release of about 14 nm per half-sarcomere.

12. When the temperature was raised both the developed tension and the stiffness increased, but the relative increase was greater for tension than for stiffness. The amount of instantaneous shortening needed to bring tension to zero was therefore also increased.

13. A set of empirical equations is given which describe adequately the first few milliseconds of the tension change in response to any imposed time course of shortening.

14. The rapid elasticity and early tension recovery resemble the response of a combination of two elastic components and one viscous component. Reasons are given for preferring an interpretation in terms of an undamped compliance in series with a damped compliance (Voigt element) rather than an undamped elasticity in parallel with a series combination of viscous and elastic components (Maxwell element).

15. The rapid compliance does not correspond to the 'series elastic component' of two-component theories of muscle contraction.

LIST OF SYMBOLS

- a radius of fibre
 $\alpha_\alpha, \alpha_\beta$, dimensionless rate constant factors determining decline of component T_α , etc., of early tension change (actual rate constant is α_α/t_1 , etc., p. 486)
 A cross-sectional area of fibre; A_r , 'reference cross-sectional area', i.e. value of A when $s = 2.1 \mu\text{m}$
 $A_\alpha A_\beta$, fractions of instantaneous stiffness associated with components etc. T_α, T_β etc. of early tension change (p. 486)

- b* damping coefficient; b_t , of force transducer (Appendix F); b_m , of fibre, due to viscosity of medium (Appendices A and C)
- c* $(C_0 + C_1)/C_t$, tendon compliance as fraction of fibre compliance (Appendices B and D)
- C* compliance; C_t , of the entire fibre; C_0 , of the attachment at the force-transducer end; C_1 , of the attachment at the motor end
- d* duration of movement of motor arm during a 'step'
- f* longitudinal force per unit area of fibre surface due to surrounding medium; f_1 , during the step; f_2 , after the step is complete (Appendices A and E)
- G* gain adjustment factor; G_L , for length signal; G_T , for tension signal
- h* reading interval for correction of force transducer output (Appendix F)
- k_t effective stiffness of force transducer (Appendix F)
- K* calibration factor for force transducer (Appendix F)
- L* distance between centres of markers
- L_{tot} total length of fibre
- m_t effective mass of moving part of force transducer
- M* total mass of fibre; $M = \rho_t A_r s_r N_{tot}$
- N* number of sarcomeres between centres of markers; $N = L/s$
- N_{tot} total number of sarcomeres in the fibre; $N_{tot} = L_{tot}/s$
- p* position of tip of motor arm (positive for displacements which increase fibre length); p_M , position indicated by internal sensor in motor; p_T , measured directly on tip of motor arm; p_s , total displacement during a step
- P* active tension per unit reference cross-section; $P = T/A_r$; P_0 , in plateau of isometric tetanus; P_e , component of P due to instantaneous stiffness and early recovery (Appendix C); P_v , component of P due to viscosity
- q* q_1 , Q_{10} of ratio (instantaneous stiffness)/(isometric tension); q_2 , Q_{10} of speed of early recovery phase
- s* sarcomere length; s_0 , preset value of s at which spot-follower output is zero; s_r , reference value of s ($= 2.1 \mu\text{m}$)
- t* time; in connexion with step responses, $t = 0$ at moment when displacement is half complete
- $t_{\frac{1}{2}}$ half-time of early recovery process as approximated by the 'standard curve'
- T* active tension, i.e. recorded tension less tension attributable to parallel elastic structures; T_0 , in plateau of an isometric tetanus; T_1 (initial value of T), immediately before applying a step; T_1 , extreme value in response to step change of length;

- T'_1 , T_1 expected on basis of linear dependence on step amplitude (p. 476); T_2 , value approached during phase 2 (p. 471); T_a , value from straight line approached during phase 2 (p. 476); T_{isot} , during isotonic phase when tetanus was started in p_M control; T_a , T_p , etc. components of early tension change (p. 486)
- u longitudinal displacement of an element of the fibre length from the position it would occupy if the fibre inertia were zero
- v longitudinal velocity (Appendices A and E)
- x displacement of moving part of force transducer (Appendix F)
- y change of length per half-sarcomere during step; $y = (s - s_0)/2$; y_s , when step is complete; y_0 , amount of instantaneous shortening per half-sarcomere (disregarding inertial effects) which would just bring T to zero
- α distance of an element of the fibre from the force-transducer end when stationary as a fraction of total fibre length
- β viscous coefficient of fibre; $\Delta P = \beta \dot{y}$
- γ initial rate constant for early tension recovery (Appendix C)
- ζ rate constant of decay of longitudinal standing waves (Appendix C)
- η viscosity (of medium surrounding the fibre)
- θ ratio of perimeter to square root of cross-sectional area
- λ space constant for decline of amplitude of fluid motion with distance from a plate oscillating in its own plane (Appendix A)
- μ complex angular frequency of longitudinal oscillations;
 $\mu = -\zeta + 2\pi i\nu_t$ (Appendix C)
- ν frequency; ν_t , of longitudinal oscillations in a fibre; ν_t , of free oscillation of force transducer
- ξ output signal of force transducer, varying linearly with displacement x of its moving part
- ρ_t density of fibre (taken as 1060 kg/m³); ρ'_t , effective value taking inertia of markers into account; ρ''_t , effective value taking markers and surrounding fluid into account (Appendix A)
- ρ_m density of surrounding medium
- σ normalized value of tension component which declines during phase 2;
 $\sigma_1 = (T - T_2)/(T_1 - T_2)$ (p. 474); $\sigma_2 = (T - T_a(t))/(T'_1 - T_a(0))$ (p. 475)
- τ time constant; τ_t , of damping of free oscillations of force transducer
- ψ function giving time course of force on fibre during step, due to viscosity and inertia of surrounding fluid (Appendix E)

INTRODUCTION

A method of investigating the mechanism of action of the cross-bridges of striated muscle is to impose a sudden change of conditions (length, load, etc.) during a contraction and to follow the resulting 'mechanical transient', i.e. the time course with which the new steady state is established. Such transients might reflect any of a wide range of processes, for example, the attachment and detachment of cross-bridges, mechanical events within cross-bridges while they are attached, elastic or viscoelastic phenomena in the filaments, and inertial effects due to displacement of the muscle mass. Some of these phenomena might be expected to occur on a time scale of a moderate number of milliseconds in frog muscle at 0 °C (A. F. Huxley, 1957) but no such transients were detected in 'quick-release' or 'controlled-release' experiments before 1960, presumably because the time resolution was not good enough (Gasser & Hill, 1924; Levin & Wyman, 1927; Hill, 1938, 1950, 1953; Jewell & Wilkie, 1958; all using whole muscles). In that year Podolsky (1960), also using whole muscles, demonstrated the existence of transients whose duration was only a few milliseconds, and Civan & Podolsky (1966) recorded similar transients from single fibres and small bundles, with a concomitant improvement in precision.

The transients studied by Podolsky and his colleagues consisted of a sequence of changes in the speed of shortening after a sudden change of load ('velocity transients') and were given an interpretation (Podolsky, Nolan & Zaveler, 1969; Podolsky & Nolan, 1973; T. L. Hill, Eisenberg, Chen & Podolsky, 1975) on a cross-bridge theory developed from that of A. F. Huxley (1957). Similar changes were also recorded in our laboratory (some records are shown in A. F. Huxley (1971)) but were not published fully because we switched to the converse type of experiment, in which the imposed change was a stepwise alteration in *length* and the time course of the resulting tension change was recorded ('tension transients'). One of the reasons for making this change of technique was that we use a servo-system which can control either the length of a fibre or the load on it according to which of the signals, length or tension, is fed back to the servo-motor. Control in the length mode is relatively straightforward, and fast responses to a step input can be achieved, but in the tension mode the feed-back signal contains the non-linear behaviour of the fibre, creating great difficulties in producing a fast, accurate change of load. Another, and more fundamental, reason was the following. We originally used step changes of load because we thought that the response would be simple to interpret: as pointed out by A. V. Hill (1938, pp. 183–184) a series elastic component would stay at constant length from the moment when the

change of load was complete, so that subsequent changes of length could all be attributed to the 'contractile component'. As our work progressed, however, it became clear that although a rapid elasticity exists, it is not 'in series' with a separable contractile component in any straightforward sense, and it now seems that interpretation is simpler when the imposed change is a step change of length.

The series of experiments that will be described and analysed in this series of papers was carried out in October 1973 to May 1974. The apparatus as a whole was similar to that described by Gordon, Huxley & Julian (1966*a, b*) but had been improved in many respects so that its time resolution was 4 or 5 times better. Experiments leading up to those described here were made by one of us (A. F. H.) in collaboration with A. M. Gordon (1960–2), F. J. Julian (1962–5) and C. M. Armstrong (1964–6) as well as R. M. Simmons (from 1967) and L. E. Ford (from 1971). Preliminary notes describing some of the results were published by Armstrong, Huxley & Julian (1966), Huxley & Simmons (1970, 1971*a, b*, 1972, 1973), and Ford, Huxley & Simmons (1974, 1976) and some other records have been published by Huxley (1971, 1974). An outline of the theory on which we interpret our results has been given by Huxley & Simmons (1971*b*, 1973).

Scope of investigation

We have concentrated mainly but not exclusively on the early phases of the response. Steps in both the shortening and the lengthening direction have been used, but the amplitude of lengthening steps has in most cases been restricted to 0.15% of the fibre length, both because of the fear that large stretches might cause permanent changes in the behaviour of the fibres and because the responses to larger stretches are complicated and deserve to be investigated separately.

Isolated fibres from the frog *Rana temporaria* were used. The anterior tibial muscle was chosen, largely because its fibres are so short (5–6 mm) that effects arising from propagation of longitudinal oscillations were not detectable. The fibres were held at 0–3 °C except when the effects of change of temperature were being investigated.

The present paper describes the methods and gives an account of the general features of the responses to step changes of length during tetani in fibres at the length where isometric tension is a maximum. The second paper in the series will describe experiments in fibres stretched to varying extents, i.e. with different amounts of overlap of thick and thin filaments, leading to the conclusion that the response is chiefly due to events within the cross-bridges themselves. Subsequent experimental papers will deal with: the response to a 'test' step applied at various time intervals after a 'conditioning' step; the response to a step superposed on a period of steady

shortening; the response to a step during the rising phase of a contraction or during relaxation. A theoretical paper will attempt an explanation of the experimental results on the basis of the theory outlined by Huxley & Simmons (1971*b*, 1973).

METHODS

Principle of the method

The fibre was held horizontal in a trough of Ringer solution. Tension was recorded with a transducer connected to one tendon; the other tendon was attached to the arm of a moving-coil device (the 'motor'). A photo-electric spot-follower continuously signalled the distance L between two markers, attached to the fibre, one near each end. The motor was servo-controlled

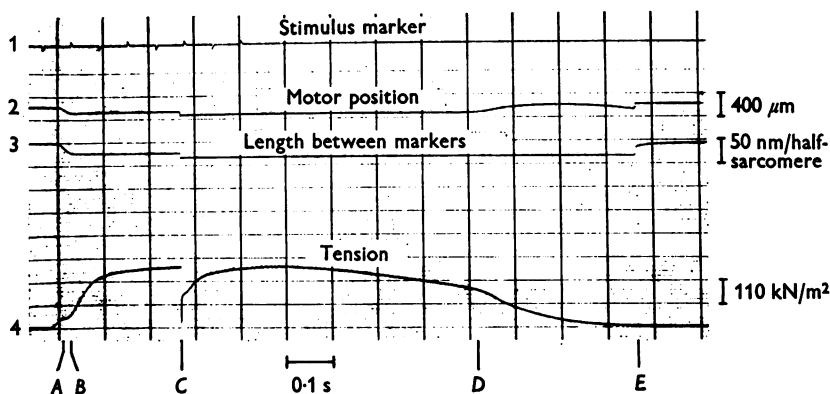


Fig. 1. Record from the U.V. recorder showing the sequence of events during a contraction. Initially the servo is controlled from the motor-position signal p_M , which therefore stays constant up to time A while the marker separation decreases very slightly because of tendon compliance. At A , tension reaches the preset level T_{isot} , and is held at that level by the servo so that the fibre shortens isotonically until B , when the sarcomere length s (proportional to the length between markers) reaches its preset level s_0 . The servo now holds s constant at this value, and the contraction is truly isometric until E , apart from the fact that a shortening step (9 nm per half-sarcomere) is imposed at C , and local elongation during relaxation begins at D . The servo is prevented from returning to tension control by a hold-off pulse which begins just before C and ends at E ; as soon as it ends, control returns to the motor-position signal. Experiment of 17 Jan. 1974; sarcomere length $2.2 \mu\text{m}$; temperature $+1.0^\circ\text{C}$.

from the spot-follower signal so as to hold L constant, or to make L undergo a predetermined change, irrespective of changes in length of the tendon or of the end regions of the fibre beyond the markers.

The motor could also be controlled from either the motor-position signal p_M or the tension signal T , and a diode circuit allowed these signals to take

over from one another as they reached pre-set levels. Control from the *L* signal was apt to be unstable when the fibre was slack because a movement of the motor in the direction to shorten the fibre was not immediately transmitted to the markers; the system was therefore usually kept in *p_M* control so long as the fibre was at rest, and passed into *L* control as the contraction developed by the sequence of events shown in Fig. 1.

Apparatus

Fig. 2 shows a block diagram of the apparatus. Schematically it was much the same as in the description given by Gordon, Huxley & Julian (1966*a*), but most parts had been rebuilt. The main changes are described in the following sections; in addition, most of the circuits had been rebuilt with solid-state in place of valve amplifiers.

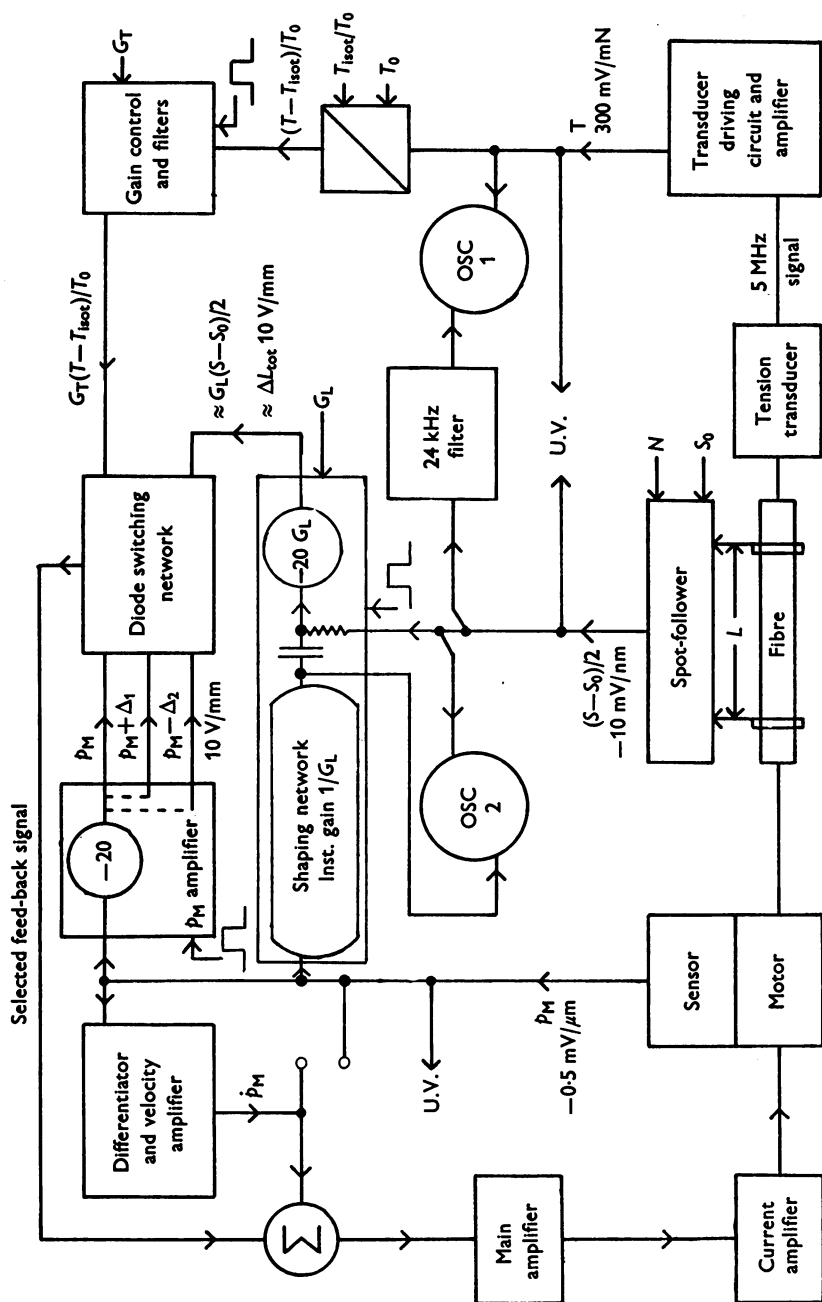
TABLE 1. Characteristics of motor and feed-back system, set up to give performance shown in Fig. 5

Length of lever arm	41 mm
Moment of inertia of moving part	$3.8 \times 10^{-7} \text{ kg.m}^2$ (= 3.8 g.cm ²)
Angular stiffness without feed-back	0.31 N.m/rad
Torque per unit current	$7.5 \times 10^{-3} \text{ N.m/A}$
DC resistance	0.8 Ω
Impedance at 3 kHz	
Total	1.23 Ω
Resistive component	1.16 Ω
Inductive component	0.46 Ω
Angular stiffness with feed-back	81 N.m/rad
Linear stiffness at hook, with feed-back	49 mN/ μm
Max. amplitude of hook displacement without saturating the current amplifier (peak output 15 V, 10 A)	70 μm

Motor. The moving-iron torque motor used by Gordon *et al.* (1966*b*) was replaced by a moving-coil device built in the laboratory, which was designed for great rigidity in its moving part so as to avoid oscillations. Its characteristics are given in Table 1.

The moving part (Fig. 3*A*) consists of a formerless coil of 50 turns of 0.5 mm enamelled aluminium wire, solidified with an epoxy adhesive, and braced by the four limbs of the pyramidal lever arm. The latter is built up from pieces of dry grass stems, coated with epoxy adhesive and reinforced with strips of carbon fibre material. The coil is pivoted by phosphor-bronze crossed-strip hinges at top and bottom, and carries a plane mirror used in the photodiode sensor (see below). Bracing was added to eliminate each mode of internal oscillation that was identified. The hook for the tendon clip was attached to the end of the arm with shellac, which could be softened by heating in order to adjust the attitude of the hook (pp. 459–460).

The pole pieces and core are shown in Fig. 3*B*. The core is sheathed with copper; eddy currents in the sheath counteract the field due to current in the coil, so that the field does not penetrate the iron, and the inductance of the coil is correspondingly



reduced; the reduction is by a factor of 2–3 at the frequencies that are important in a 'step' movement of the motor (we are indebted to Professor O. R. Frisch for suggesting the use of copper sheathing). The magnetic field is provided by a permanent horseshoe magnet (Mullard M 4403), giving a field in the gap of 4500 oersted. Current is passed through the coil by an amplifier ('current amplifier', Fig. 2), which uses an internal feed-back loop to make its output current proportional to its input voltage, thus eliminating the lag that would otherwise be caused by the inductance of the coil. The amplifier was fed from independent power supplies with large capacitors across the outputs to provide the surges (up to ± 10 A) needed during the steps; this arrangement also avoided any possibility of large steady currents which might overheat the motor.

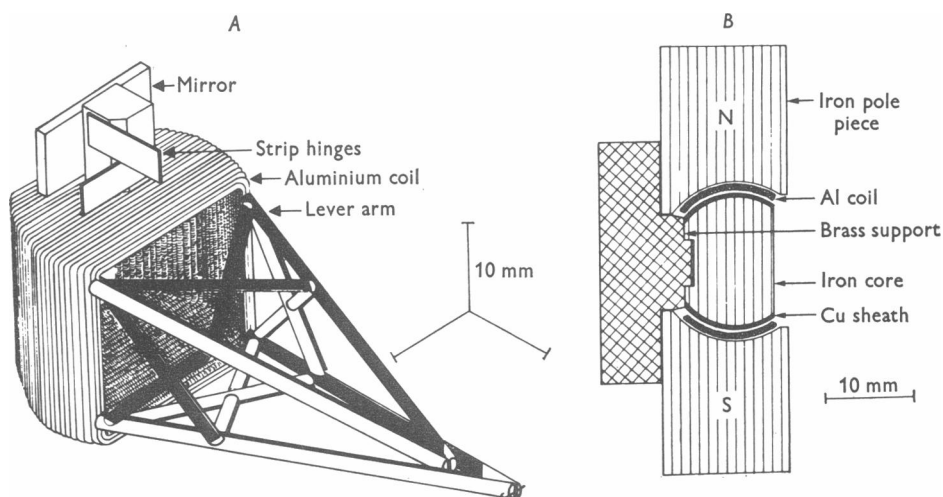


Fig. 3. Construction of the moving-coil motor. *A*, the moving part (isometric view). A number of bracing struts, added as required to eliminate particular modes of vibration, are not shown. *B*, cross-section through pole pieces, core and coil.

Rotation of the moving part is signalled by a photodiode sensor (Fig. 4). The source is a 6 V, 6 W bulb with a closely wound filament, underrun at 3.6 V for stability. The optical system is designed so that the amount of light falling on each of the photodiodes varies linearly with rotation of the coil together with the mirror about its proper axis but is unaffected by any other components of motion that the mirror may undergo as a result of bending within the moving part or by movement of the filament.

The photodiodes (Mullard, BPX 42) are connected as shown in Fig. 4*B* so that the difference between their (reverse) currents is fed to an operational amplifier. The 'sensitivity' resistor (approx. 6 k Ω) was adjusted before each experiment so that the output was 0.5 V per mm displacement of the hook, measured with a microscope; the setting barely altered from one day to another. The output followed an instantaneous change of current with a nearly exponential lag of about 2.5 μ s.

The whole motor was mounted on a rack-and-pinion slide with vernier scale for rough pre-setting of fibre length.

Spot-follower. The spot-follower, which gives a signal proportional to changes in

the distance between two markers stuck to the fibre, was the device described by Gordon *et al.* (1966*a*) with the following changes:

(a) A shorter-focal-length lens (Kern-Paillard 'Switar' 25 mm, f 1.5) was used and the cathode-ray tube was moved further from the lens, so that the optical reduction was $10\times$ instead of $3\times$. This improved the signal-to-noise ratio in direct proportion since a given change of marker separation corresponded to a larger displacement of the spots and therefore gave a larger signal.

(b) An aperture stop was placed at the lower focal plane of the lens so that the axial ray of each beam was vertical as it passed through the trough. This made the output insensitive to up-and-down movements of the fibre.

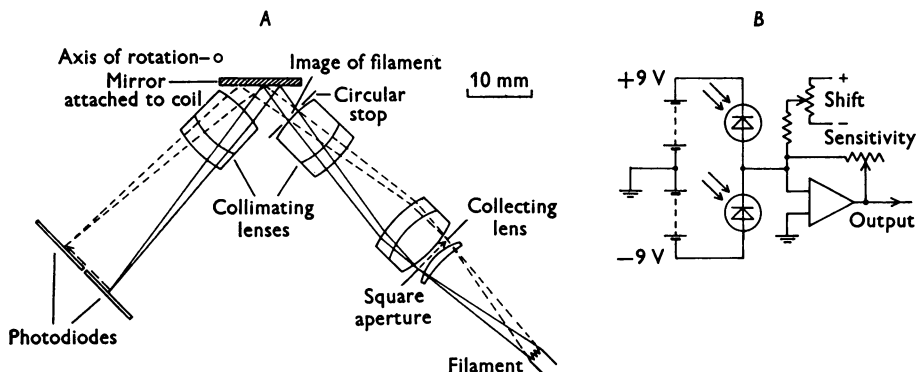


Fig. 4. Photodiode sensor on motor. *A*, diagram of optical layout, roughly to scale. The collecting lens is the optical part of a Zeiss $\times 25$ eyepiece for a stereo microscope. The other two lenses are the optical parts of Beck 32 mm microscope objectives. The mirror is placed off centre relative to the axis of rotation in order that the reflected light shall still pass through the middle of the final lens even when the mirror is deflected. *B*, simplified diagram of the control circuit for the photodiodes.

(c) Each spot on the C.R.T. was drawn out into a line approximately at right angles to the fibre axis by a triangular waveform at 24 kHz. The directions of the lines could be adjusted electrically so as to bring each exactly parallel to the edge of the corresponding marker. The adjustment could be made while the spot-follower was operating, by observing X - Y displays of the photocell outputs against the deflecting voltage. Final adjustments were made by watching these X - Y displays during a twitch or tetanus. Residual 24 kHz noise on the output was removed from the signal recorded on oscilloscope 1 by a notch filter but the unfiltered signal was also recorded (with a Polaroid camera) from oscilloscope 2 (see Fig. 2). This filter did not introduce detectable distortions.

(d) The scale of the output signal was made equal to 10 mV for 1 nm length change per half-sarcomere by incorporating a 10-turn potentiometer whose dial was set to N (the number of sarcomeres between the mid-points of the markers), 1 turn representing 1000 sarcomeres. The pulse amplitude could therefore be set directly in nm per half-sarcomere.

(e) The spot intensity was increased, giving a photocurrent of 10^{-9} A in each photocell when the spot-follower was in operation. To avoid burning of the tube face, a switch was provided which disconnected the feed-back and deflected the spots to another part of the tube except for a short period at each contraction.

The image of each spot at the fibre level was a line 0.3–0.4 mm long. The intensity distribution across its width was roughly Gaussian with standard deviation 10 μm .

The performance of the spot-follower is illustrated in Fig. 5*A*. The lowest trace, a differential recording between the spot-follower output and a signal proportional to p_M , when both are signalling displacement of the motor arm, shows hardly any deflexion. There is a barely detectable diphasic deflexion during the motor movement, a lag during acceleration and a lead during retardation of the motor, i.e. in the same direction as the deviation of tip movement (recorded with photodiodes, Fig. 5*B*) from the p_M signal. The spot-follower signal has much more noise than the p_M signal.

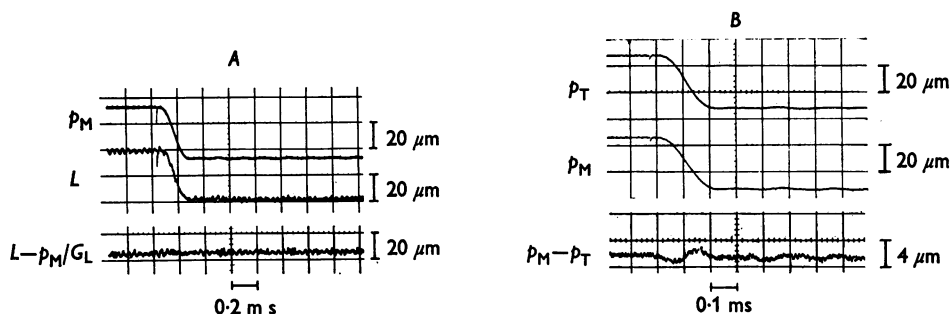


Fig. 5. *A*, performance of spot-follower. Upper trace, motor position p_M signalled by the internal sensor; middle trace, spot-follower signal L recorded through the notch filter; bottom trace, a differential record showing $L - p_M/G_L$, G_L having been adjusted to eliminate any net displacement between the early and late parts of the trace. The left-hand spot of the spot-follower is following the motor arm near its tip, while the right-hand spot is locked on to a stationary 'dummy marker'. *B*, performance of servo. Upper trace, p_T , i.e. position of tip of motor arm signalled by a pair of photodiodes adjusted so that amplitude of step change of voltage is equal to that in the middle trace; middle trace, p_M , i.e. motor position signalled by the internal sensor; bottom trace, differential record of the quantities recorded separately in the upper and middle traces (note higher gain).

Feed-back loop. The feed-back signal fed to the current amplifier consisted of a 'direct' and a 'velocity' component. The direct component came either from the photodiode sensor on the motor (p_M signal), or from the spot-follower (s or L), or from the force transducer (T), one of these three being selected by the diode switching circuit (see below). The velocity component was always derived from p_M , since the L signal was too noisy and T was affected by the transducer characteristics and was subject to non-linearities which varied with the state of the fibre. The p_M signal was differentiated by a circuit which deliberately introduced two successive first-order lags whose values (both 8 μs) were chosen so as to give a step that was complete as early as possible.

In each experiment the servo was initially adjusted while performing steps in 'motor control', i.e. with direct feed-back from the p_M signal. The command signal was a ramp-and-hold pulse, with ramp duration 80 μs . The gain of the 'main amplifier' (Fig. 2) was set so that the underdamped frequency was 2.33 kHz, and the gain

of the 'velocity amplifier' was then increased until the p_M signal had the form shown in Fig. 5. The settings obtained by this procedure were then left unaltered throughout the experiment or occasionally readjusted by repeating the same procedure, while the performance when the direct component came from the spot-follower signal ('length control') or from tension ('tension control') was adjusted by setting the gain factor G_L or G_T (Fig. 2) respectively. In the present series of experiments, tension control was used only for the isotonic phase at the start of a tetanus (A-B, Fig. 1) before the spot-follower signal took control, so the factor G_T was not critical. The value of G_L was checked by examining the record of the spot-follower signal obtained from an oscilloscope (OSC 2, Fig. 2) with a 'Polaroid' camera after each contraction in which a step in length control was made: if the initial step change was larger or smaller than the steady-state level, the value of G_L was decreased or increased respectively and a fresh record taken in the next contraction.

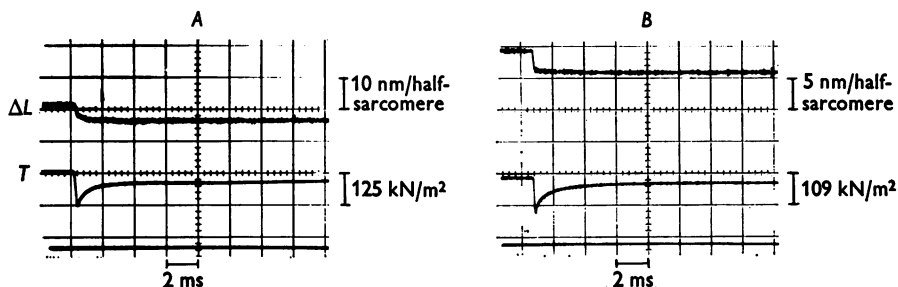


Fig. 6. Reduction of tendon compliance by means of tendon clips. Shortening imposed by step displacement of motor tip, without feedback from spot-follower. *A*, fibre held by hooks inserted through the tendons (expt. of 11 June 1973; 4.7 °C). *B*, tendons held by metal clips as shown in Fig. 8*A* (expt. of 17 Jan. 1974; 1.0 °C). Upper trace: distance between markers, signalled by spot-follower (note different scales in *A* and *B*). Middle trace: tension. Lower trace: resting tension base line. In *A*, the shortening during the first 3–4 ms after completion of the step amounts to 1.8 nm per half-sarcomere (0.16 % of length) as the tension recovers from 0.57 P_0 to 0.85 P_0 ; the external compliance is therefore $0.16/(0.85-0.57) = 0.57\% L_0/P_0$. In *B*, the corresponding figures are 0.05 nm, 0.49 P_0 and 0.92 P_0 , so the external compliance has been reduced to 0.11 % L_0/P_0 .

The noise on the spot-follower input to the diode switching network was reduced by condenser coupling (time constant 0.25 ms) from a signal derived from p_M (Fig. 2). This signal was shaped by an R-C network so that it should not introduce a spurious component due to the motor movements that were necessary, during the first 1 or 2 ms after completion of the step itself, to re-stretch the tendon compliance as the tension in the fibre rose. These tension changes varied in a non-linear manner with step amplitude, so the parameters of the shaping network had to be readjusted for each size of the step. Adjustments were made on the basis of characteristic deviations from the ideal step form in the Polaroid camera record of the spot-follower signal. The number of repetitions needed in order to obtain a satisfactory step increases rapidly with the amount of compliance in the tendons, and was a limiting factor in preliminary experiments when the tendons were held simply by putting hooks through them; after we began using metal clips to hold the tendons, however (p. 459; Fig. 8*A*), the settings could usually be estimated in advance and it was very

seldom necessary to make more than one readjustment for each step size. Fig. 6 shows that the tendon compliance was reduced about fivefold by using the clips.

Servo performance. The motion of the tip of the motor arm, p_T , differed slightly from that signalled by the mirror built into the motor, p_M , as shown in Fig. 5 *B*. It was fairly well approximated during the rising phase ($-\frac{1}{2}d < t < \frac{1}{2}d$) by the formula

$$p_T = p_s (1 + \sin \pi t/d)/2, \quad (1)$$

where p_s is the displacement when the step is complete, t is time measured from the moment when the step is half complete (about 130 μ s after the timing marks) and d , the total duration of the movement, is 200 μ s.

Shorter rise times could be achieved by increasing the loop gain but only at the cost of a rapid increase in the amplitude of the oscillations at about 5 kHz which can be seen after the step in Figs. 5 and 10. These are related to a resonance in the motor, but we did not succeed in identifying the mode of flexure responsible for this oscillation.

Occasionally irregular oscillations at 12–16 kHz were seen on the tension record, often before the step was applied (see, for example, the left-hand middle record in Fig. 14). These did not originate in the spot-follower, since they were seen even when the servo was operating in motor control. Their frequency is higher than the very well-defined damped oscillatory response of the tension transducer. Their frequency is near the value expected for longitudinal oscillations in the fibre (about 16 kHz, see Appendix B), but we believe they do not originate in this way because

(a) they were seen when the fibre was replaced by a filament of terylene or glass, which would have frequencies of longitudinal oscillation 5 or 10 times higher;

(b) their frequency was not lower in stretched fibres, whereas the reduced stiffness due to reduced overlap would cause a large drop in the frequency of longitudinal oscillations in a fibre (Appendix B).

They are almost certainly due to oscillations in the motor arm. The motor showed several resonances between 12 and 16 kHz, but the modes of flexure responsible for them were not identified.

With feed-back from either p_M or the spot-follower signal, the restoring force per unit displacement at the hook was 49 mN/ μ m (Table 1). This figure is some 300 times greater than either the stiffness due to the hinges (0.19 mN/ μ m) or the instantaneous stiffness of the muscle fibre (a typical tetanic tension of 5 mN, dropping to zero in 0.5% of the fibre length (6 mm), corresponds to a stiffness of 0.17 mN/ μ m). Errors from these sources were therefore small, and no steps were taken to reduce them further.

Diode switching network. This circuit, shown diagrammatically in Fig. 7, operates on the same principles as that used by Gordon *et al.* (1966*a, b*) but has an additional input and is connected so as to produce a different sequence of control. The diodes operate in the manner described by Gordon *et al.* (1966*a*, p. 153) with the over-all result that the output potential is equal to whichever is lowest of the potentials of the following three of the inputs:

- (1) whichever of inputs *A* and *B* has the higher potential,
- (2) whichever of inputs *C* and *D* has the higher potential,
- (3) input *E*.

Inputs *C* and *E* (Fig. 7), at potentials respectively below and above that of the p_M signal by preset amounts, provide limits to the range within which the motor can be driven by inputs *B* and *D*; their main purpose was to protect the fibre against the violent motor movement which would otherwise be caused if (as occasionally happened) one of the spots of the spot-follower became accidentally displaced across its marker, producing a very large signal at input *D*.

The sequence of events in a contraction such as is shown in Fig. 1 is set out in Table 2.

Force transducer. Tension in the fibre was signalled by a transducer of the capacitance-gauge type described by Huxley & Simmons (1968). Its construction is shown in Fig. 8B. It is built from pieces of glass stuck together with epoxy resin (Araldite, CIBA) diluted with benzene to give a very thin layer. The base is a piece of 4.5 mm plate glass. Three pieces of microscope slide are stuck to the base; the central piece, which is slightly thinner than the others, is metallized by vacuum deposition of gold and acts as the fixed plate of the capacitor. The outer pieces of slide support a strip of 0.2 mm cover-glass which is metallized in its middle part and acts as the moving plate. The gap between the plates (20 μm) is set by the difference in thickness between the pieces of microscope slide. The air in the gap provides damping.

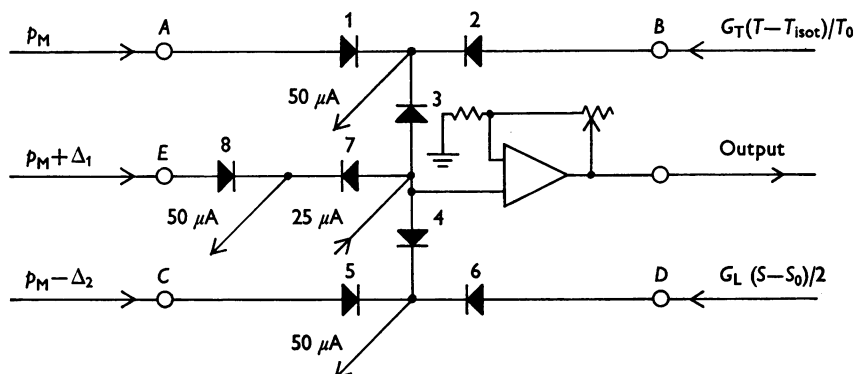


Fig. 7. Simplified diagram of diode switching network. For mode of operation, see page 454. Differences between individual diodes are corrected for by adjusting the nominal 50 μA currents, and by presetting resistors (not shown) in series with diodes 1, 2, 5, 6 and 8. The amplifier has high input impedance and its gain (slightly above unity) is adjusted so that the overall gain from any one input to the output is exactly unity. The diodes are RS components ISJ150 silicon junction diodes.

TABLE 2. Operation of diode switching network (Fig. 7) during contraction shown in Fig. 1

Stage of contraction (Fig. 1)	Control signal, held at zero by the servo action	Diodes through which control signal is connected to output	Diodes which are non- conducting
Start, until A	p_M (input A)	1, 3	2, 4, 5, 7
A-B	$(T - T_{\text{isot}})$ (input B)	2, 3	1, 4, 5, 7
B-E	$(s - s_0)$ (input D)	4, 6	1, 3, 5, 7
After E	p_M (input A)	1, 3	2, 4, 5, 7

From shortly before C until E (Fig. 1) a positive voltage is added to the signal at input B, to prevent the system from returning to tension control when the shortening step is applied at time C and the tension falls. In contractions where $T > T_{\text{isot}}$ at time E (end of the 'tension hold-off'), control passes from $(s - s_0)$ through $(T - T_{\text{isot}})$ to p_M , reversing the sequence of events at the start of the contraction.

Connexions to the plates are made with thin copper wires stuck to the pieces of glass before they were metallized.

Changes of capacitance caused by forces applied to the moving plate were converted to DC signals by the phase-discriminator circuit, working at a fixed frequency of 5 MHz, described by Cambridge & Haines (1959) and Haines (1961). The output followed an instantaneous capacitance change with a roughly exponential lag of about 5 μ s.

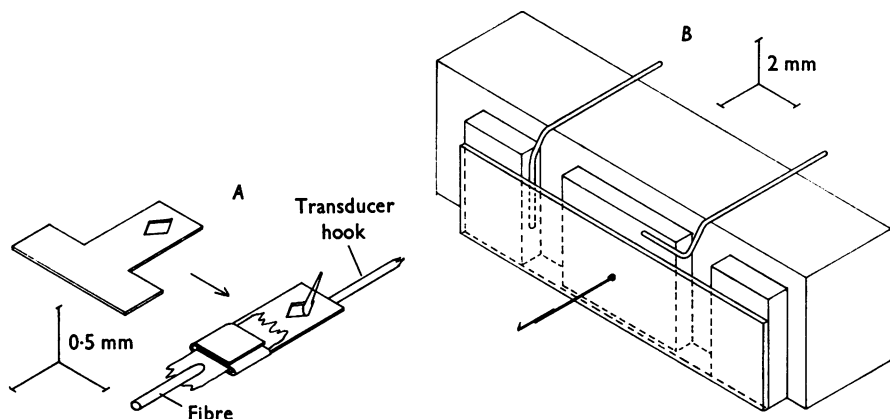


Fig. 8. *A*, tendon clips, to scale. They are made from 25 μ m Al foil and weigh approx. 55 μ g each. *B*, construction of force transducer. For description, see text. To scale, except that the gap, set by the difference in thickness between the central and the two outer pieces of glass, is drawn 5 times too wide. The shield is not shown, nor are the metallized areas on the plates.

The tension from the fibre was applied directly to the middle of the moving element. A small pin was stuck to its centre and to this the hook for the tendon clip was attached with shellac. The pin protruded horizontally into the Ringer solution through a notch cut in the end of the trough, which was greased to prevent the solution from escaping. The attitude of the hook could be altered by melting the shellac (see pp. 459–460).

To protect the transducer against accidental flooding, and to reduce thermal drift, a shield was made to cover the front of the instrument except for a small hole for the pin.

The performance of the transducer is illustrated in Fig. 9. There is a short-lived damped oscillation at 10.8 kHz when the transducer is suddenly unloaded, but no appreciable overshoot or lag in approaching the steady signal. The characteristics of the instrument are given in Table 3.

Fig. 10 shows the response of the transducer to a change of force whose time-course is similar to those recorded in our experiments. A glass filament, which can be assumed to have ideal elastic properties, was glued between the motor tip and the transducer hook. The displacement of the motor tip was recorded so that if the transducer performance had been perfect the two traces would have had exactly similar time courses. The transducer signal shows an initial lag of some 15 μ s and an overshoot of about 3%, which are due chiefly to the inertia of the moving part of the transducer. The circles show the output corrected for inertia and damping by the procedure described in Appendix F. The result is almost exactly proportional to

motor displacement but is delayed by about $1.4 \mu\text{s}$; this is attributable to the lag in the circuits associated with the transducer (p. 456) being greater than that in the photodiode circuits (p. 450).

The transducer was calibrated at the end of each experiment as described by Gordon *et al.* (1966a).

The particular transducer used in these experiments suffered from thermal drift, probably due to differential expansion effects between the adhesive and the glass. However, the drift was negligible over the time course of a single contraction. A tension base line was recorded on every oscilloscope picture immediately before the contraction.

Trough. The trough was mounted in a stainless-steel plate which also carried adjustable slides for the motor, the force transducer, and one of the stimulating

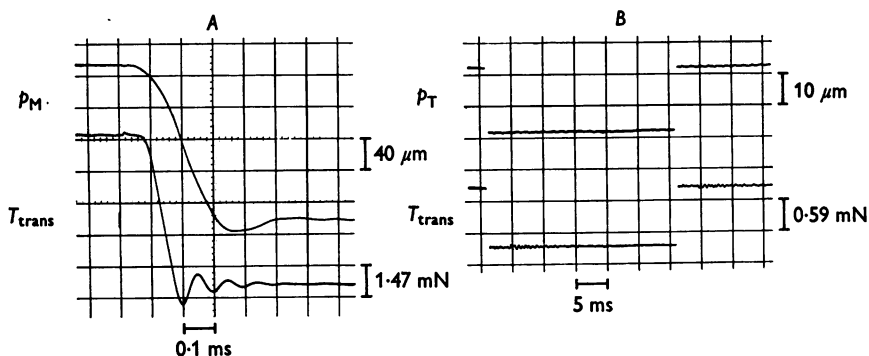


Fig. 9. Transducer performance. *A*, lower trace shows transducer output signal when the transducer is suddenly unloaded. A polyester filament ($10 \mu\text{m}$ diam.) with a loop at each end connects the hooks on the motor arm and on the transducer, and the motor is made to perform a large step towards the transducer so that the thread goes slack and the loop loses contact with the transducer hook. From a time just before the first minimum in the output signal, the transducer is completely unloaded and performs a damped oscillation about the level corresponding to zero tension (the overshoot on the motor displacement (upper trace) is due to the step being so large that the current amplifier saturates).

B, records on a slow time base to show the absence of overshoot or creep in the transducer output in times of the order of 1–20 ms. The motor arm and the transducer are connected by a glass filament (approx. 14 mm long, $4 \mu\text{m}$ diam.) attached at each end with shellac. The motor makes a step in p_M control with amplitude about three-quarters of that needed to bring the tension in the filament to zero. Upper trace, displacement of tip of motor arm, detected with a pair of photodiodes; lower trace, transducer output signal.

TABLE 3. Characteristics of force transducer (typical isometric tension was 5 mN)

Sensitivity	50 mV/mN
Natural frequency	10.8 kHz
Damping time constant	0.10 ms
Compliance	0.03 $\mu\text{m}/\text{mN}$
Effective mass	8 mg

electrodes. The plate could be fitted interchangeably to the dissecting microscope or to the mechanical stage of either the microscope stand on which the experiment was performed or another microscope stand on which visual measurements of fibre diameter, striation spacing, marker position and fibre length were made.

Below the glass bottom of the trough were two thin chambers formed by additional sheets of glass. Cooling fluid circulated in the upper of these chambers; the lower contained air dried with silica gel to prevent condensation. Temperatures in the trough were not measured during each experiment but were estimated from the temperature of the cooling fluid and the room temperature, on the basis of measurements

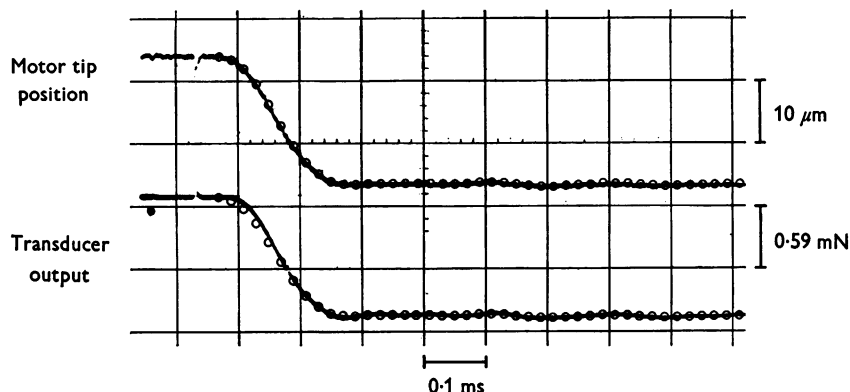


Fig. 10. Correction of transducer signal for inertia and damping in the transducer. Conditions same as in Fig. 9*B*, except for the higher time-base speed. C.R.T. traces: upper, motor tip position; lower, transducer output signal. Circles on lower trace: transducer output signal read every 20 μ s and corrected by the procedure described in Appendix F. Circles on upper trace: same, but multiplied by a constant factor to bring them into coincidence with the trace in the later part of the record.

with a thermocouple placed at the position normally occupied by the fibre. The temperatures stated are the values at the middle of the fibre; owing to the proximity of the end of the trough there was a temperature gradient along the fibre amounting to about 1 $^{\circ}$ C in the region between the markers.

Timing. Two 'Digitimers' (Devices Ltd.) provided timing pulses for the start and end of stimulation, for the start and end of command pulses and the 'tension hold-off pulse', and for triggering the oscilloscopes. The Digitimers and the U.V. recorder were started by a contact on the camera of oscilloscope 1.

Recording

All amplifiers in the recording systems were direct coupled.

Three records were normally taken in each contraction.

(1) A 35 mm photograph of OSC 1 (Fig. 2), which was a Tektronix 502 oscilloscope. Time-base speeds from 0.2 to 2 ms/cm were used. This was the record from which the early events in the transients were analysed. Timing pulses coincident with the start of the command pulse were applied to both traces through the differential inputs. The grid lines (but not the actual traces) have in some cases been re-touched for reproduction in this paper.

2. A Polaroid photograph of OSC 2, which was a Solartron CD 1400 oscilloscope. This record of L and p_M/G_L , usually at a time-base speed of 0.5 ms/cm, was used only for checking and correcting the adjustments of the servo (p. 453).

(3) A record on an U.V. galvanometer recorder (S.E. Laboratories, model 2006), usually at a paper speed of 20 cm/s; Fig. 1 is an example. The record was used for measuring the slower components of the transient responses and, during the experiment, for checking the performance of the fibre and for tracing the cause of accidents. The natural frequencies of the galvanometers used were 5 kHz for spot-follower and force transducer outputs, and 2.5 kHz for the p_M and stimulus signals. Steep parts of the traces have been retouched for reproduction.

Preparation of fibres

Dissection. Fibres were dissected as described by Gordon *et al.* (1966a) except that m. tibialis anterior (lateral head) was used, not semitendinosus. Tibialis anterior was chosen (cf. Sten-Knudsen, 1953) because (a) its fibres are shorter (see Discussion, p. 504), (b) the tendinous sheets into which its fibres insert are more solid, and (c) its fibres are more elliptical in section, allowing better contact with the markers. Fibres were selected for large size (major diam. $\approx 200\ \mu\text{m}$ at slack length) and for firm attachment to tendon. Fibres were tested by tetanic stimulation at every stage of preparation. The pieces of tendon left attached at each end were carefully trimmed to length 0.7 mm, breadth 0.4 mm approx. The fibre was then transferred from the dissecting dish to the experimental trough.

Tendon clips. The tendon pieces were clamped in metal clips (Fig. 8A) which were bent tightly over the tendon with one edge within about 0.2 mm of the end of the fibre. Each clip had a hole to receive the hook attached to motor or transducer. At this stage of the preparation, however, they were placed on 'dummy hooks', which were rotated around the fibre axis so as to eliminate twists and to bring the major axis of the fibre cross-section horizontal.

Markers. Gold leaf markers attached as described by Gordon *et al.* (1966a) tended to move relative to the fibre when subjected to the large accelerations (up to 500 g) that accompanied the steps used in the present experiments. The mass of each marker was therefore reduced from 30–40 μg to about 2 μg by using 3 μm titanium foil in place of gold and reducing the dimensions to 0.6–0.7 mm by 0.2–0.3 mm. To bring the centre of mass close to the fibre axis, and to improve adhesion to the fibre, the markers were bent into a V shape with included angle approx. 120° ; the fibre lay along the bottom of the V. The markers were stuck to the fibre with tap grease (Stevens's Rubber Grease) as described by Gordon *et al.* (1966a); in cold weather the Ringer solution was warmed to 25°C to soften the grease. The attachment of the markers was done with the fibre stretched beyond the maximum length to be used in the experiment so that the fibre would expand into the ring of grease when it was shortened, which was not done until the trough was cooled to near 0°C . With the same purpose, a hypertonic solution ($1.5\times$ Ringer) was used in some experiments while the markers were being attached.

The behaviour of the markers was checked in a few experiments by watching them under stroboscopic illumination with a high-power microscope ($\times 40$ water-immersion objective) while undergoing step changes of length. A successfully attached marker showed no detectable lag or lead relative to the striations. In some cases, relative movement was seen when the fibre was stretched to, say, $3.0\ \mu\text{m}$ sarcomere length for experiments to be described in a later paper, but the markers behaved satisfactorily at shorter lengths.

Adjustment of hooks. The fibre was stimulated tetanically, and if the markers did not stay horizontal the dummy hooks were rotated so that the markers came into

the horizontal position on stimulation. The hooks on the motor and transducer were adjusted to the same attitude as the corresponding dummy by softening the shellac which held them. After completing the measurements, the motor and the force transducer were attached to the stage plate and the clips transferred to their hooks without allowing the fibre to shorten, and the dummies were removed.

Measurements. The stage plate was transferred from the dissecting microscope to an ordinary high-power microscope fitted with a dial gauge for precise measurement of movement of the plate in the direction of the fibre axis. The positions of the fibre ends and the right-hand edges of the markers were determined with the dial gauge, giving total length of fibre and length between the markers. With a $\times 40$ water-immersion objective and a $\times 25$ micrometer eyepiece, the following measurements were made visually at each of four positions between the markers.

(a) Two counts, in different parts of the field, of the number of striations (to the nearest 0.2) in 20 divisions of the micrometer ($54\ \mu\text{m}$ approx.).

(b) Extreme width of the fibre.

(c) Vertical height of the fibre measured with the fine focusing motion, at the middle of the fibre width.

The mean of the eight sarcomere counts was taken and converted to sarcomere length s ; dividing this into the marker separation gave N , which was set on the appropriate dial of the spot-follower. Cross-sectional area was taken as $\frac{1}{4}\pi$ times the product of the measurements (b) and (c); multiplying this by $s/2.1\ \mu\text{m}$ gave A_r , the cross-sectional area at the reference sarcomere length of $2.1\ \mu\text{m}$.

Adjustment of spot-follower, etc. The stage plate was next transferred to the microscope on which the spot-follower is built. The circulation of cooling fluid was begun. The fibre was shortened to the required length by shifting the motor. Threshold and fusion frequency were measured. The spots were focused to the level of the markers. The fibre was shifted so that the spot images lay across it, final adjustments to the spot positions being made electrically. The directions of the lines formed by the spots were adjusted to be parallel to the marker edges (see p. 451). The photocells were mounted above the fibre and the feed-back in the spot-follower turned on. The X - Y displays of photocell output were watched during contractions, and final adjustments were made so as to bring the lines parallel to the marker edges. A trial tetanus with shortening step in p_m control was recorded, and the experiment was then begun. Tetani were given at regular intervals of 4–7 min. Most of the fibres lasted for 20–30 h after completion of the setting-up procedure, and during this time around 200 tetani were recorded.

Stimulation

The fibre was stimulated by pulses of transverse current, of alternating polarity, and with amplitude about 1.5 times threshold.

Pulses from a Devices 'gated pulse generator' set to the required stimulus frequency triggered a valve flip-flop circuit whose output was fed through a potentiometer and condenser to the primary of a step-down transformer. The output was applied to a pair of blacked-platinum electrodes $4 \times 12\ \text{mm}$, about 4 mm apart, one on each side of the fibre. One of the electrodes was carried on a screw-operated slide so that it could be placed exactly opposite the other, thus ensuring that the current was accurately transverse.

The stimulus wave form was roughly triangular, falling to zero in about 2 ms and overshooting slightly. The peak voltage gradient in the trough was of the order of 20 V/cm, varying according to the size of the fibre.

The timing of the stimuli was marked by recording the primary current on the U.V. recorder.

Solutions

The composition of the Ringer solution was 115 mM-NaCl, 2.5 mM-KCl, 1.8 mM- CaCl_2 , 3.0 mM phosphate buffer pH 7.0 (Adrian, 1956). D-tubocurarine (10^{-5} g/ml) was added to all solutions. Ringer solution was made up freshly for each experiment from solid NaCl and KCl and stock solutions of CaCl_2 and buffer, which were kept at 4 °C. The Ringer solution in the trough was changed at intervals during the experiment because it was slowly diluted by condensation.

Corrections

When a detailed comparison was to be made between the time course of the imposed length change and that of the resulting tension change, it was necessary to apply the following corrections to the output of the force transducer.

(a) *Correction for inertia and damping in the force transducer.* The theory of the correction procedure is given in Appendix F. The record of transducer output on OSC 1 was measured at intervals of 20 μs or occasionally 10 μs and corrected according to eqn. (F 14) or (F 15) respectively.

(b) *Effect of fibre inertia.* Longitudinal oscillations were not evident in the records (p. 454) but Appendix D shows that part of the force needed to accelerate the mass of the fibre appears at the force transducer. This was calculated from eqn. (D 10), assuming the time course of change of length of the fibre to be similar to that of the movement of the motor tip (Fig. 5B), and was subtracted from the force indicated by the transducer.

(c) *Effect of surrounding fluid.* The combined effect, during the step, of inertia and viscosity in the Ringer solution surrounding the fibre is calculated in Appendix E. The resulting force on the transducer was evaluated from eqn. (E 7) with $\theta = 4$, and subtracted from the force indicated by the transducer.

RESULTS

General features of the tension transients

Fig. 1 shows the whole of a tetanus with a shortening step applied during the plateau. It illustrates the process by which the servo is brought into the 'length-control' mode at the start of the contraction as explained in the Methods section (pp. 454–455).

Fig. 11 shows the tension traces from the family of records which includes Fig. 1. The earliest parts of the tension changes in the shortening steps are so rapid that they appear only as a vertical line at this speed; after this, and before the final, roughly exponential, recovery towards the isometric tension there is a period when the rate of rise of tension is greatly reduced or even reversed. Even in the smallest release, when the early phase of tension recovery brings the tension almost back to the isometric value, the later phases are evident in the form of a slight fall of tension followed by slow recovery. Similarly, in response to the lengthening step, a period of rising tension is interposed between the early rapid fall of tension and the final decay towards the isometric value.

Fig. 12 shows oscilloscope records of the early parts of a similar family

of transients, taken at a sweep speed of 2 ms/cm. In each tension trace there is an initial step at the time of the imposed length change, after which the tension recovers, in a few milliseconds, part of the way towards the isometric value. The extreme tension T_1 reached in the initial step, and the tension T_2 that is approached during this early recovery phase,

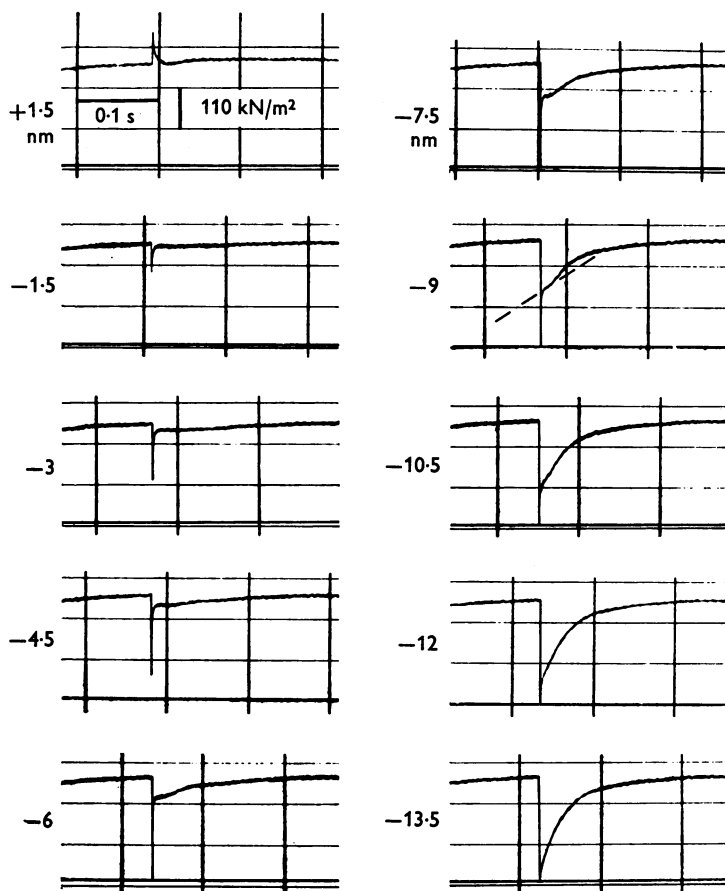


Fig. 11. U.V. recorder traces of tension during tetani with step change of length imposed 0.3 s after first stimulus. Amplitude of step per half-sarcomere given opposite each record. Motor controlled by feed-back from spot-follower signal. Fig. 1 shows the whole of the contraction with 9 nm shortening per half-sarcomere in this family. The broken line on the 9 nm record is the tangent to the tension trace at the point of inflexion during phase 3; its intersection with the initial drop of tension is taken as T_2 (p. 471). Expt. of 17 Jan. 1974; sarcomere length $2.2 \mu\text{m}$; temperature $+1.0^\circ\text{C}$. Steep part of each record, and horizontal grid lines, retouched; heavy horizontal line has been added and indicates the tension base line.

are plotted against the size of the step in Fig. 13. Inspection of the records in Fig. 12 shows that the early recovery is slowest in the response to the larger stretch, and becomes progressively faster from the smallest to the largest shortening response (excluding the 9 nm and 12 nm releases, in which the step was intentionally slowed).

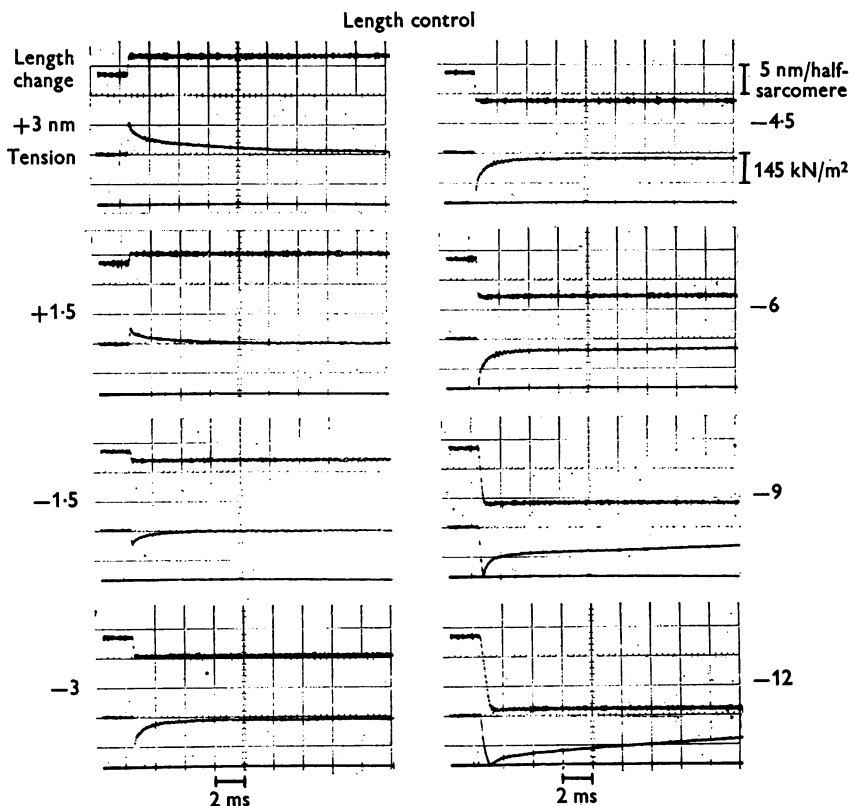


Fig. 12. Early phases of tension change in response to step change of length during a tetanus. Motor controlled by feed-back from spot-follower signal. Upper trace: marker separation, from spot-follower. Middle trace: tension. Bottom trace: base line for tension, recorded immediately before the contraction. Amplitude of length change per half-sarcomere given alongside each panel in nm. In the 9 nm and 12 nm releases, the step was deliberately slowed so as to prevent the fibre from going slack. In the other cases the standard step of 0.2 ms total duration was used. Expt. of 2 April 1974; sarcomere length $2.2 \mu\text{m}$; temperature $+2.5^\circ\text{C}$.

Fig. 14 shows another family of transients on a still faster time-base (0.2 ms/cm). It can now be seen that the initial tension change develops with a time course not very different from that of the imposed length change.

The tension changes that are seen in Figs. 11, 12 and 14 during and after a step can thus be divided into the four phases set out in Table 4. As has already been pointed out (Huxley, 1971, 1974; Huxley & Simmons, 1973), phases that correspond to these can also be recognized in the converse type of experiment, where the tension is made to change in a stepwise manner and the time course of the length change ('velocity transient') is recorded (except in phase 1, however, the time scale of a particular phase in the velocity transient is not simply related to that of the corresponding phase in the tension transient).

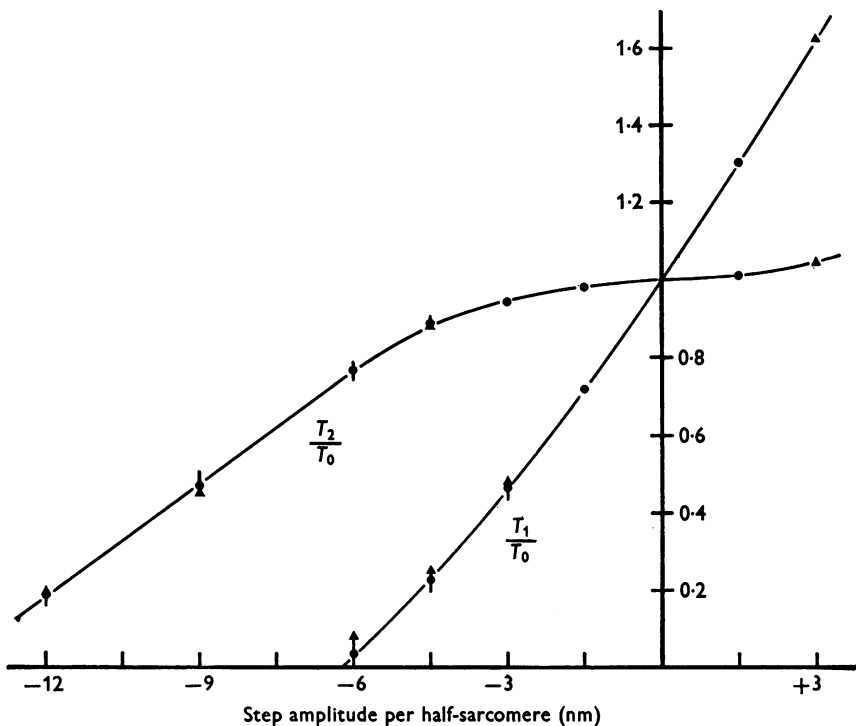


Fig. 13. Curves of T_1 (extreme tension) and T_2 (tension approached during early recovery phase) in length-control steps of various amplitudes, both expressed as fractions of T_0 , the isometric tension immediately before the step. T_2 estimated as described on page 471. Expt. of 2 April 1974. Circles, means, and vertical lines, range, of first four series at sarcomere length $2.2 \mu\text{m}$. Triangles: fourth of these series, from records reproduced in Fig. 12; where no triangle is shown the measurement coincided with the corresponding circle. Temperature $+2.5^\circ\text{C}$.

A few features of phases 4 and 3 will be mentioned on pages 470 and 471, but a more thorough examination of them requires consideration of the double-step experiments that will be described in a later paper of this

series. Most of the detailed results to be reported in the present paper have to do with phase 2 (p. 471) and phase 1 (p. 480).

Responses without spot-follower feed-back. The records in Figs. 12 and 14 were obtained with the servo operating so as to regulate the distance

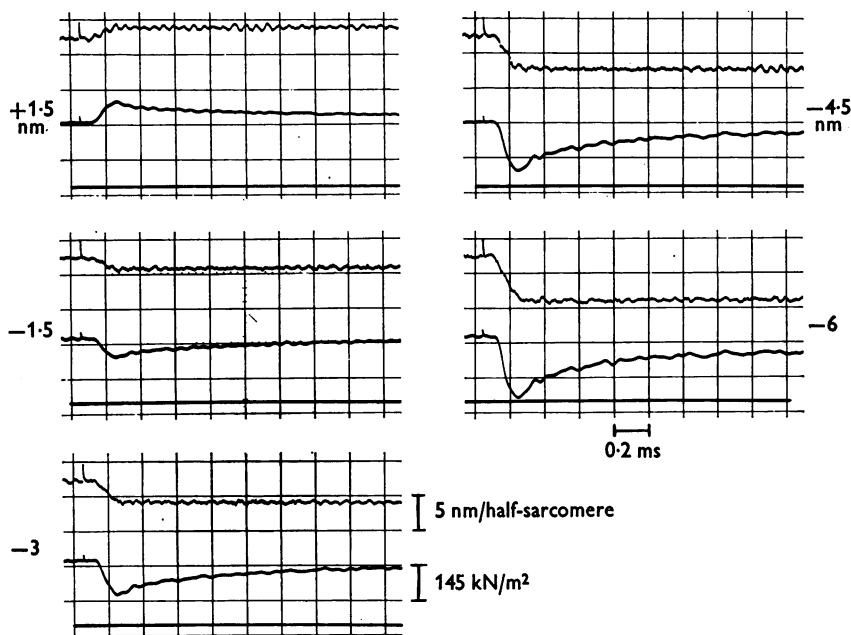


Fig. 14. As Fig. 12, but on 10 times faster time-base, to show that initial tension change (middle trace) occurs almost synchronously with the length change recorded by the spot-follower (upper trace). Amplitude of length change per half-sarcomere given alongside each panel, in nm. Expt. of 2 April 1974; sarcomere length $2.0 \mu\text{m}$; temperature $+2.5^\circ\text{C}$.

TABLE 4. Phases of the transient response to sudden reduction of length ('tension transient') or of load ('velocity transient'), in frog muscle at 0°C

Phase	Time of occurrence	Event in tension transient	Event in velocity transient
1	During applied step change	Simultaneous drop of tension	Simultaneous shortening
2	Next 2-5 ms	Rapid early tension recovery	Rapid early shortening
3	Next 10-50 ms	Extreme reduction or even reversal of rate of tension recovery	Extreme reduction or even reversal of rate of shortening
4	Remainder of the response	Gradual recovery of tension, with asymptotic approach to isometric tension	Shortening at steady speed, sometimes with superposed damped oscillation

between the markers signalled by the spot-follower. Any suspicion that major features of the tension changes might be due to imperfections in the operation of this rather complex feed-back system is removed by records such as are shown in Fig. 15, which were obtained with the motor

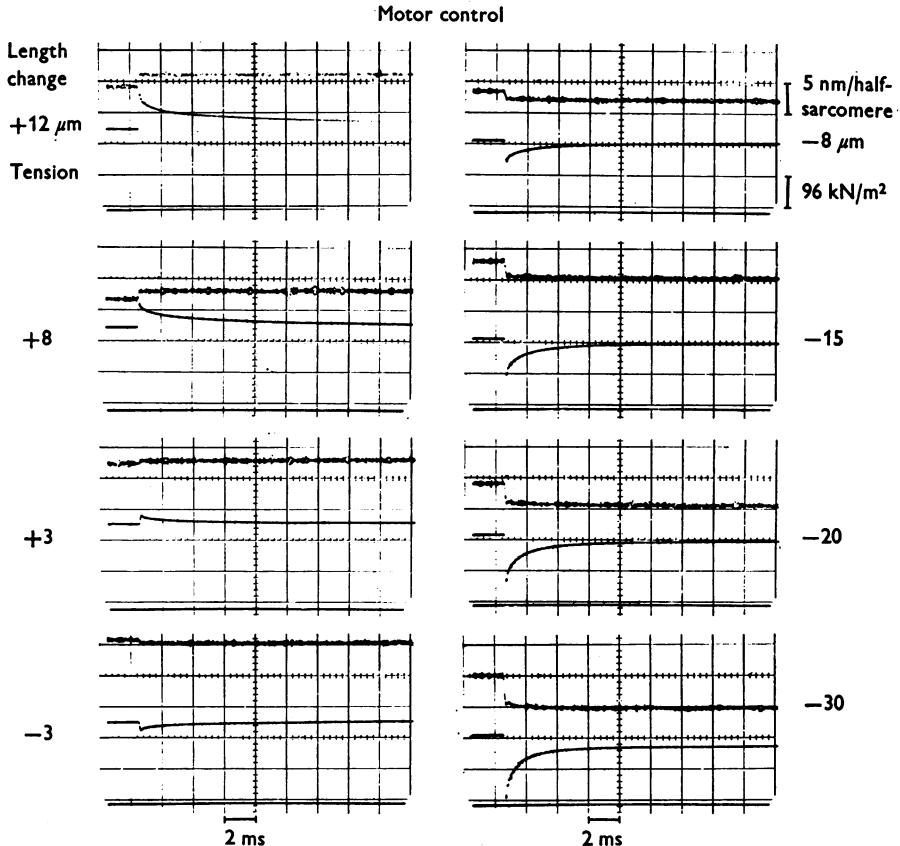


Fig. 15. Responses to step displacements, the motor being controlled by feed-back from its internal position signal and not from the spot-follower output. Top trace: distance between markers, signalled by spot-follower. Middle trace: tension. Bottom trace: base line for tension. Amplitude of step (in μm) given opposite each panel. Expt. of 5 Dec. 1973; sarcomere length $2.2 \mu\text{m}$; total number of sarcomeres in the fibre approx. 2700; temperature $+1.0^\circ\text{C}$.

arm undergoing a step displacement in 'motor control' (p. 452; Fig. 5*B*), i.e. without feed-back from the spot-follower signal. The tension changes do not differ in any qualitative way from those shown in Fig. 12, where the spot-follower feedback was in use. There is, however, in the larger releases a noticeable downward creep in the spot-follower signal after the step,

which is due to extension of the tendon attachments, and therefore shortening of the fibre itself, as the tension rises, so that the contractions were not truly isometric.

Fig. 16 presents a direct comparison between records made under 'length control' and 'motor control'. The effect of the 'creep' on the tension time course is evident in the lower pair of records. Because of the creep the change of length of the fibre (indicated by the spot-follower signal) in motor control is smaller immediately after the step, but larger at times greater than 1 ms after the step, than in the length-control record. As a result T_1/T_0 is noticeably larger and T_2/T_0 is smaller in the motor-control than in the length-control record.

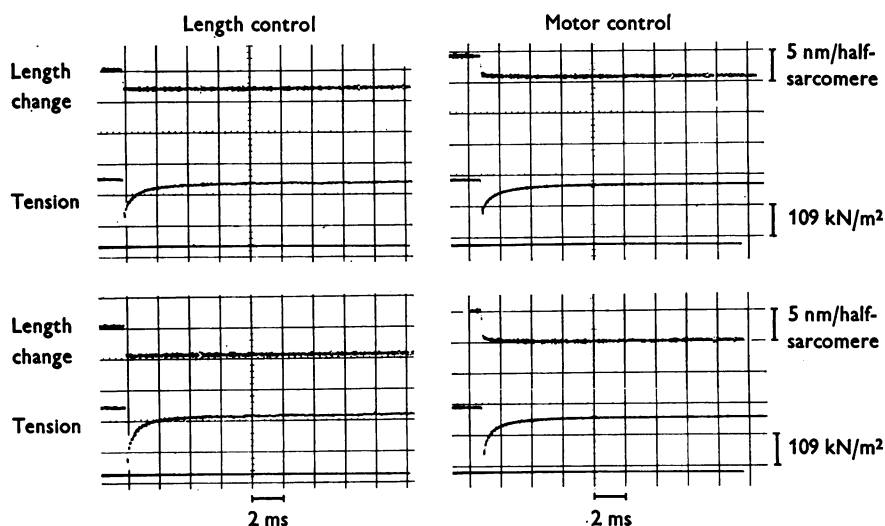


Fig. 16. Step changes imposed with motor controlled from spot-follower signal (left) and from its internal position signal (right). Amplitudes: left upper, 3 nm per half-sarcomere; left lower, 4.5 nm per half-sarcomere; right upper, 20 μ m displacement of tendon; right lower, 30 μ m displacement of tendon. Upper traces: spot-follower output. From consecutive series in the experiment of 17 Jan. 1974. Sarcomere length 2.2 μ m; total number of sarcomeres in the fibre approx. 2800; temperature +1.0 °C.

Because of these distortions of the tension time course in motor control, all the quantitative analysis in this paper is based on records made in the length-control mode. The motor-control records do, however, have the advantage that the noise on the tension traces is smaller than when length control is used: in the latter case the noise on the spot-follower signal gives rise to small unwanted movements of the motor arm which in turn give rise to changes of tension. Motor-control records are therefore used in the comparison between small stretches and releases (pp. 474–475; Fig. 21).

The distortion of the tension time course when spot-follower feed-back was not employed was much more severe in preliminary experiments before we began using metal clips to grip the tendons, and the tendons were held simply by inserting hooks (Fig. 6).

Stiffness and viscosity in unstimulated fibres

Some part of the tension transients in stimulated fibres may be attributable to passive structures in parallel with the contractile material. It is therefore necessary to know the response of an unstimulated fibre to a step change of length. This could not be recorded directly in a fibre near its slack length where the resting tension is practically zero; in a shortening step there is no detectable tension change, and in a step stretch the

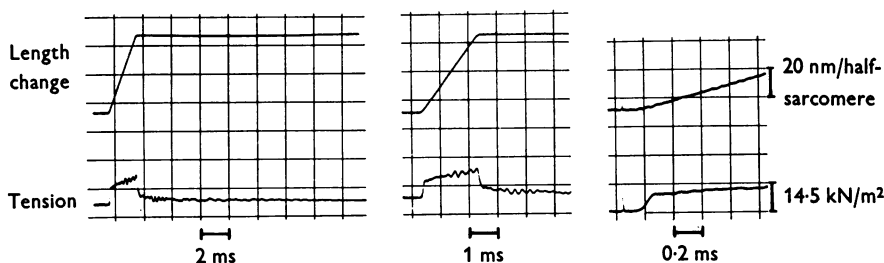


Fig. 17. Tension changes due to stretch of an unstimulated muscle fibre. The fibre is subjected to a 0.3 mm ramp stretch of duration 2 ms in 'motor control', i.e. feed-back is from the motor sensor signal p_M . The upper trace, however, is the output of the spot-follower. The three records differ only in the time-base speed. Expt. of 2 April 1974; sarcomere length 2.2 μm ; temperature +2.5 °C.

tension is not sufficient to prevent the fibre from overshooting at the end of the step, the tendon clip failing to remain in contact with the motor hook as the latter decelerates. The passive properties of the fibres were therefore investigated by means of ramp stretches at speeds comparable to those attained during the steps. As shown in Fig. 17, the tension rises almost synchronously with the rate of elongation, and falls again when the ramp ends; when a lower ramp speed was used the tension changes were reduced approximately in proportion to the speed. These aspects of the behaviour correspond to a 'viscosity' in the fibre. A viscous coefficient β can be estimated by dividing the rate of stretching of each half-sarcomere (about 27 nm/ms) into the force per unit cross-section (about 8 kN/m²), giving $\beta = 3 \times 10^8 \text{ N.s/m}^3$. Similar measurements in other experiments gave values of β ranging from 1.5×10^8 to $4 \times 10^8 \text{ N.s/m}^3$. These values show a strong and highly significant negative correlation with fibre cross-section; indeed, the total force when subjected to a given rate of stretch was not obviously correlated with cross-sectional area.

In addition to this 'viscous' behaviour, there is also a progressive rise of tension throughout the ramp but this extra tension falls gradually after the ramp is ended. The tension has almost reached its original level at the end of the slowest of the records (1 kN/m^2 at 16 ms after the end of the ramp), and is then still falling. This agrees with other measurements which show that, in most fibres, resting tension caused by a stretch of this magnitude, and measured at times of the order of minutes, is less than 0.5 kN/m^2 (e.g. Ramsey & Street, 1940; L  nnergren, 1971; Rapoport, 1972).

The slow rise of tension during the ramp in Fig. 17, and the tension decay after it, are well fitted by postulating that any elongation Δy per half sarcomere at time t_1 produces tension change ΔP (in addition to that attributable to 'viscosity') given by the equation ($t > t_1$):

$$\Delta P = \{a_1 \exp -(t-t_1)/\tau_1 + a_2 \exp -(t-t_1)/\tau_2\} \Delta y, \quad (2)$$

with $a_1 = 19 \times 10^{10} \text{ N/m}^3$, $\tau_1 = 0.65 \text{ ms}$, $a_2 = 4.6 \times 10^{10} \text{ N/m}^3$, $\tau_2 = 20 \text{ ms}$.

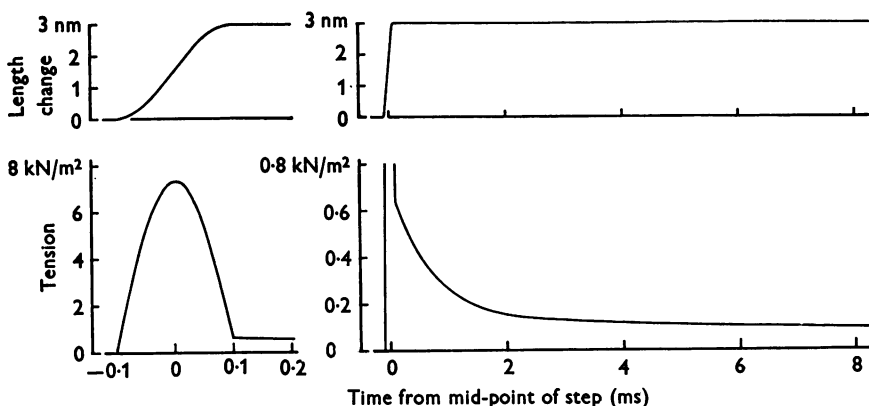


Fig. 18. Calculated passive tension changes in response to 'step' length change of amplitude 3 nm per half-sarcomere with the form described by equation (1) ($d = 0.2 \text{ ms}$). Viscous coefficient β taken as $3.0 \times 10^8 \text{ N.s/m}^3$; after-effects as described by eqn. (2).

The form of eqn. (2) is convenient for calculating by superposition the tension changes during and after a length change of any time course. Taking eqn. (1) (p. 454) as an approximation to the form of the steps applied to stimulated fibres, and assuming ideal properties in the viscosity itself, the resulting tension time course is shown in Fig. 18.

It is not immediately clear whether the apparent 'viscosity' resembles a true viscosity in giving a resistive force proportional to instantaneous rate of elongation, or whether it is a stiffness with short but appreciable

relaxation time. In principle it is possible to estimate the relaxation time by measuring the delay between the onset of the motor movement and the mid-point of the tension rise, since the theoretical value of this delay contains a component roughly equal to the relaxation time. However, the delay is sensitive to compliance and viscosity in the tendon attachments and these are not accurately known. Tendon compliance can be estimated roughly from the lag between motor movement (p_M signal) and spot-follower signal (40–50 μs in the experiment of Fig. 17), and approximate calculations suggest that the observed delay (about 80 μs) is fully accounted for without assuming an appreciable value for the relaxation time. On this basis it seems likely that any relaxation time is less than perhaps 20 μs and may well be very much shorter, but further experiments are needed before a firm conclusion can be reached.

Phase 4 of the transient (final tension recovery)

The time course of this final stage of recovery is seen in the records of Fig. 11, for a series of steps of different amplitude. The last part ($T > 0.8T_0$) of this stage of the recovery after the larger shortening steps usually has a time course indistinguishable from that of the rise of tension over the same range at the start of the tetanus, but the early part of the recovery after a large release is more rapid than the corresponding part of the initial tension development. This result is similar to the finding of Jewell & Wilkie (1958), although the experimental conditions are different in that those authors gave a relatively large release (approx. 6% of muscle length) so that the muscle was shortening freely for some time before the redevelopment of tension began.

The time courses of phase 4 after releases of different amplitude cannot be made to superpose exactly on each other by any combination of scale changes and horizontal or vertical shifts, except that, as mentioned above, the final approach to the isometric tension is similar in all cases.

The speed of phase 4 is markedly increased by a rise of temperature, the effect being comparable to that on the speed of the initial rise of tension. We have not investigated the effect systematically; in one experiment releases of 9 nm per half-sarcomere were made at four temperatures ranging from 2.5 to 21.3 °C, and the speed of phase 4 increased with a Q_{10} of about 2.5.

There is no suggestion that the value of tension that was approached asymptotically after the step length change was different from that in an isometric tetanus. The point was not firmly established, however, since the experiments were not designed for testing it, and in most of them the tetanus duration was kept short in the hope of reducing fatigue, with the result that relaxation began before tension recovery was complete.

Phase 3 of the transient (slowing or reversal of recovery)

This phase is much more conspicuous in some preparations than in others; we have not found the cause of this variability. In stretches, in small releases, and sometimes in the larger releases, it shows up as an actual reversal of the rate of change of tension. Its duration is greatest in large stretches, and smallest in large releases, and appears to vary continuously in between, with no suggestion of a discontinuity between small stretches and small releases. In experiments where runs were performed at more than one temperature, this phase was much more conspicuous at 7 °C and above than in the neighbourhood of 0 °C.

Phase 2 of the transient (early tension recovery)

Much of the tension change that occurs synchronously with the length step is reversed during the next few milliseconds, as seen in Figs. 12 and 14. This behaviour is qualitatively like that of an undamped spring in series with a damped spring (Voigt element), but it shows non-linearities as regards both its extent and its speed.

Extent of early recovery

The first of these non-linearities is illustrated in Fig. 13, in which the tension T_2 approached during this phase is plotted against the step amplitude. It is seen that the recovery is almost complete in stretches and releases in which the length change in each half-sarcomere is only a few nanometres, but becomes progressively less complete in the larger releases so that the curve is concave downward.

The exact definition of T_2 is to some extent arbitrary since the physical basis of phase 3 of the transient, which follows the early tension recovery, is not yet definitely known, and there is therefore no reliable basis for extrapolating backwards the slow changes of tension upon which the early tension recovery is superposed. At this stage we use the most conservative procedure, as follows.

(1) *Stretches.* The tension always passes through a minimum, and this minimum is taken as T_2 .

(2) *Releases in which tension passes through a maximum.* The maximum is taken as T_2 .

(3) *Larger releases.* In all cases there is a point of inflexion in the time course of tension during phase 3. A tangent is drawn at this point (see 9 nm release in Fig. 11) and its intersection with the initial fall of tension is taken as T_2 .

T_2 and the shape of the step. The curve of T_2 against step amplitude in Fig. 13 is very similar to that obtained (Huxley & Simmons, 1971a) with the earlier version of our equipment in which the step was complete in about 1 ms. In one experiment of the present series, the step was deliberately slowed to match the earlier apparatus, and the values of T_1 and T_2

obtained with both 0.2 ms and 1 ms steps are plotted in Fig. 19. The T_2 curves are indistinguishable, although the T_1 curves differ substantially (see page 489).

In other experiments (to be reported in detail in a later paper of this series), two shortening steps were applied at an interval of 1–2 ms. Table 5 shows the T_2 level reached after the second of such a pair of steps, compared with that reached after a single step of the same total amplitude. It is seen that the values of T_2 are closely similar.

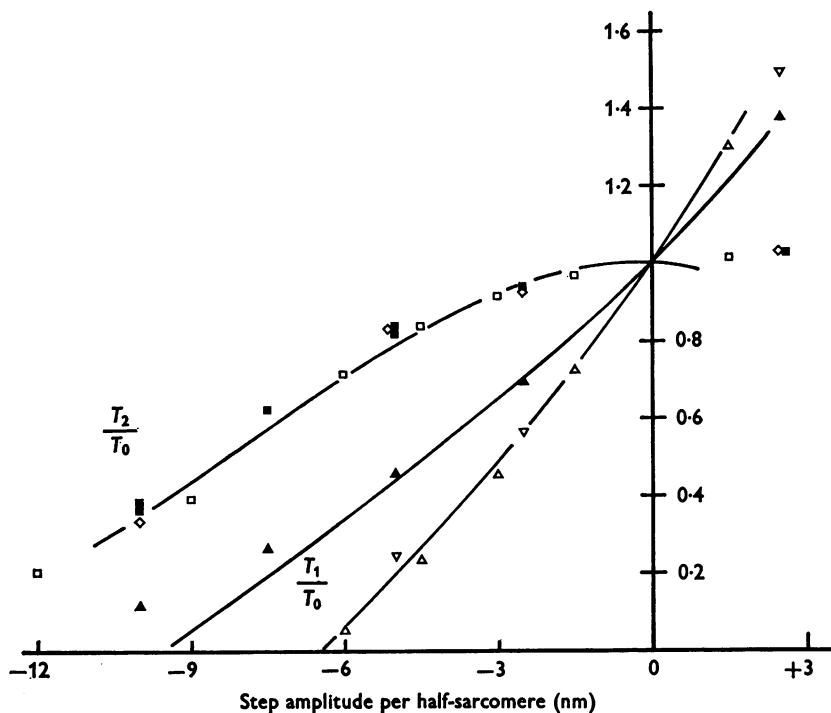


Fig. 19. Curves of T_1 (triangles) and T_2 (squares) obtained with the standard step complete in 0.2 ms (open symbols), and with a step complete in 1.0 ms (filled symbols). Three consecutive series: (1) $\triangle$, T_1 , and $\square$, T_2 , with 0.2 ms step; (2) $\blacktriangle$, T_1 , and $\blacksquare$, T_2 , with 1.0 ms step; (3) ∇ , T_1 , and $\diamond$, T_2 , with 0.2 ms step. Upper line: cubic fitted to T_2 values in the shortening steps, and used in the computer simulation (p. 489). Lower lines: computed T_1/T_0 for 0.2 ms step (bottom) and 1 ms step (middle). Expt. of 30 Oct. 1973; sarcomere length $2.1 \mu\text{m}$; temperature $+1.3^\circ\text{C}$.

Thus, the T_2 level is determined almost entirely by the total amplitude of the step change of length and is not sensitive to its time course, provided it is complete within 2 ms. In particular, the extreme tension reached during the step can vary widely without appreciable change in T_2 .

TABLE 5. Values of T_2/T_0 after single and double shortening steps. In each row, total amount of shortening is constant. y_1 , y_2 , step amplitudes per half-sarcomere for first step and for second step respectively. Interval between first and second steps, 2 ms

Expt.	Total shortening $-(y_1 + y_2)$	Single step T_2/T_0	Double steps		
			$y_1 = -1.5$ nm T_2/T_0	$y_1 = -3$ nm T_2/T_0	$y_1 = -4.5$ nm T_2/T_0
23. x. 73	4.5 nm	0.864	0.862	0.855	—
	6.0 nm	0.739	0.744	0.759	0.748
5. xii. 73	3.0 nm	0.928	0.924	—	—
	4.5 nm	0.878	0.884	0.878	—
	6.0 nm	0.777	0.781	0.783	0.782

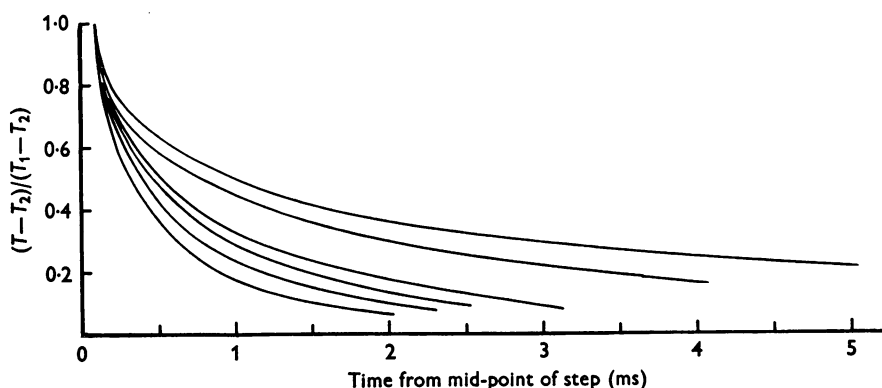


Fig. 20. Time course of early recovery phase of tension following steps of various amplitudes. Re-plotted from tracings of the records reproduced in Fig. 12. Step amplitudes per half-sarcomere, from top curve to bottom curve: +3 nm, +1.5 nm, -1.5 nm, -3 nm, -4.5 nm, -6 nm. The quantity plotted is $\sigma_1 = (T - T_2)/(T_1 - T_2)$, where T_1 is the extreme tension reached and T_2 is the tension approached in the early recovery phase, according to definitions on page 471. This quantity necessarily has the value 1 at the peak of the tension change since T is then equal to T_1 . The peak occurs about 0.1 ms after the middle of the imposed step change of length. Expt. of 2 April 1974; sarcomere length 2.2 μ m; temperature +2.5 $^{\circ}$ C.

Dependence of T_2 on isometric tension. There was little if any change in the curve of T_2/T_0 against step amplitude during the course of an experiment when runs under similar conditions were compared, although the absolute value of T_0 fluctuated appreciably (e.g. when the interval between contractions was changed) and also fell by some 25–30 % during the later stages of a 20 h experiment. The value of T_2 for a given size of step thus varied in direct proportion to the isometric tension.

The T_2 curve did, however, change appreciably when the isometric tension rose as a result of raising the temperature (see pages 492 *et seq.*).

Speed of early tension recovery

The second non-linearity in the early tension recovery is the fact that its time course varies with the direction and amplitude of the step, being slowest in a large stretch, fastest in a large release, and varying continuously in between. This is evident from inspection of Figs. 12, 14 and 15, but is brought out more clearly in Fig. 20, where the early recovery phases are superposed after scaling up to the same amplitude.

The quantity σ_1 plotted as ordinate in Fig. 20 is defined by

$$\sigma_1 = (T - T_2)/(T_1 - T_2), \quad (3)$$

where T_1 is the extreme tension reached and T_2 is measured according to the criteria on page 471. This is the simplest quantity that can be used as a measure of the component of tension which undergoes this early decay; it under-estimates the effect of step size on the speed of the decay in the following two ways.

(1) The absolute value of the denominator is under-estimated because of the effect of the limited speed of the step on the value of T_1 . This effect is more severe the greater the speed of the early recovery; hence the lower curves in Fig. 20 need to be reduced in vertical scale relative to the upper ones, reducing further the time required for decay to any given level.

(2) The numerator should ideally be the deviation of T from some line that is approached asymptotically, rather than from the constant T_2 . The time courses calculated in this way would be slower than those shown in Fig. 20, and the effect is greatest in the stretches and smallest, in this case negligible, in the large releases.

A different kind of phenomenon which also causes the effect to be under-estimated in Fig. 20 is that the speed of the early recovery phase in a stretch was found to become progressively greater during the course of an experiment. For example, the 1.5 nm stretch in Fig. 20 was applied during the 117th contraction by that fibre, and the time course was 3–4 times faster than in the 9th contraction when a stretch of the same amplitude was given (see Fig. 22). In the case of shortening steps, progressive changes of this kind were of a smaller order of magnitude; in several cases they were not detectable and the largest effect (in the experiment of 5 Feb. 1974) was an acceleration of the time scale of the recoveries by a factor of 1.25 in the course of 175 contractions. The reciprocal of the half-time increased by roughly equal amounts in steps of different amplitude.

Continuity between stretches and releases. In Fig. 20 there is a large gap between the smallest stretch and the smallest release. In several

experiments smaller steps (± 0.75 nm per half-sarcomere) were applied and the rates of recovery were intermediate between those for $+1.5$ and -1.5 nm per half-sarcomere, but the deflexions were too small in relation to the noise to allow the time course to be measured adequately. As mentioned earlier, however (p. 467), the noise on the tension record was much less when the motor was servo-controlled from its internal position signal (motor control) than when it was controlled from the spot-follower. The series of responses of this kind shown in Fig. 15 includes steps of about ± 0.5 and ± 1.4 nm per half-sarcomere; these are re-plotted on a logarithmic scale in Fig. 21, and they show no discontinuity in rate of the early recovery phase between stretching and shortening steps.

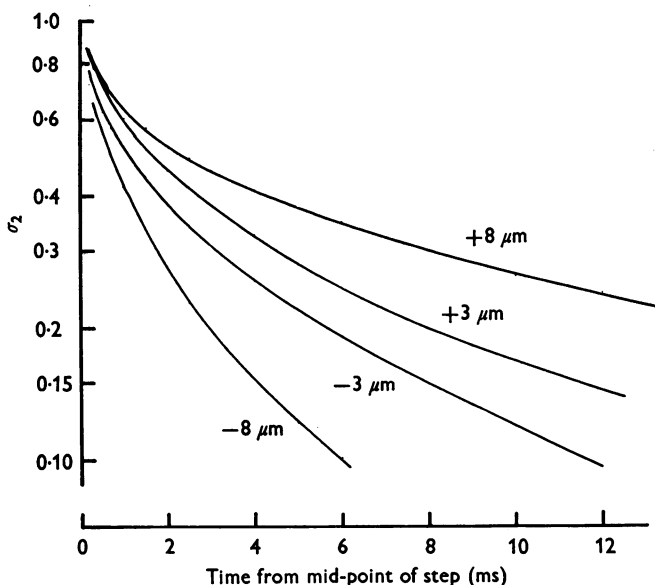


Fig. 21. Early tension recovery following small step displacements, to show continuity between stretches and releases. Motor controlled by feed-back from its internal position signal. The figure by each curve gives the amplitude of the displacement step applied to the tendon; $1 \mu\text{m}$ displacement corresponds to about 0.18 nm length change per half-sarcomere. Quantity plotted is σ_2 , on logarithmic scale. From records reproduced in Fig. 15.

Time course of early tension recovery. The time courses plotted in Fig. 20 are subject to the two sources of error mentioned on page 474. The same data are re-plotted in Fig. 22, on a logarithmic scale, after re-calculation to make approximate allowance for these errors. The quantity plotted denoted by σ_2 , is defined by the equation

$$\sigma_2 = (T - T_a(t)) / (T'_1 - T_a(0)), \quad (4)$$

where (i) T'_1 , which replaces T_1 in eqn. (3), is taken as a linear function of the step amplitude, on a scale such that in the largest stretch of the series, it is equal to the measured T_1 , and

(ii) T_s , which replaces T_2 in eqn. (3), is the ordinate of the straight line which is approached by the tension during the early recovery phase. This line was taken to be the tangent at the point of inflexion during phase 3, drawn by eye on the low-speed record (see Fig. 11).

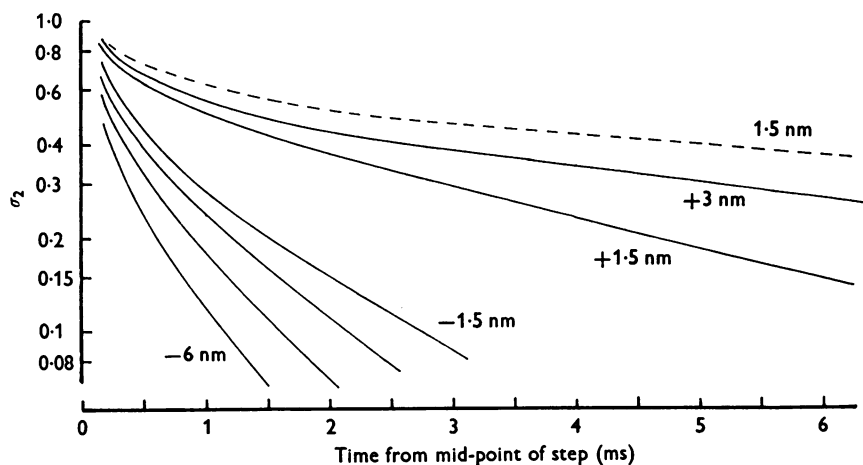


Fig. 22. Early recovery phases plotted semilogarithmically. Continuous lines: same data as Fig. 20 but quantity plotted is σ_2 (p. 475); from contractions nos. 113–118 in experiment of 2 April 1974. Interrupted line: 1.5 nm stretch in contraction no. 9 in the same experiment. Curves from shortening steps in contractions between no. 6 and no. 11 are only slightly slower than the continuous lines in this Figure.

The first of these corrections is equivalent to assuming that the instantaneous elasticity obeys Hooke's law; the validity of this assumption will be examined on page 488. The only effect of this correction on the curves in Fig. 22 is to displace them downwards, without change of shape, by amounts which increase from zero in the 3 nm stretch to a maximum in the 6 nm release.

The earliest measurement of tension used in constructing these curves is at $t = 0.15$ ms, i.e. about 0.05 ms after the completion of the step. Tension during the step itself is affected by several complicating factors, discussed on pages 481 *et seq.*

The use of a logarithmic scale in Fig. 22 brings out the fact, already evident in Figs. 12 and 20, that the time course is not exponential; this is another deviation from the behaviour of a Voigt element composed of ideal elastic and viscous structures, in addition to the non-linearity of the

T_2 curve (p. 471) and the variation of the speed of the early tension recovery with step amplitude (p. 474). Each curve can of course be fitted by a sum of several exponential components, and this procedure is useful in providing a numerical description of the form of the curves (see below and pp. 479, 485, and 493). The actual values of the amplitudes and rates of the components are, however, of limited significance because of the latitude of choice in the fitting process.

Huxley & Simmons (1972) found that the curves then available could be fitted adequately by taking a single (non-exponential) decay curve and adjusting its time scale. We have tried out this procedure by replotting the values of σ_2 from Fig. 22, and similar data from the other experiments, on double logarithmic coordinates, and we found that the responses to shortening steps, but not the responses to stretches, can be fitted adequately by shifting a single curve along parallel to the time axis. If, however, the early part of this 'standard' curve from the shortening responses is matched to the corresponding part of a response to a stretch, then the later part of the standard curve declines too quickly. The deviations are slight in cases where the time course is relatively rapid (e.g. the 1.5 nm stretch in Fig. 20) but are very conspicuous when it is slow (e.g. the interrupted line in Fig. 22).

The 'standard' curve, which represents approximately the early recovery phase after an instantaneous step, is

$$\begin{aligned} \sigma_2 = & 0.2 \exp [-0.13 t/t_{\frac{1}{2}}] + 0.45 \exp [-0.47 t/t_{\frac{1}{2}}] \\ & + 0.2 \exp [-1.60 t/t_{\frac{1}{2}}] + 0.3 \exp [-6.9 t/t_{\frac{1}{2}}], \end{aligned} \quad (5)$$

where $t_{\frac{1}{2}}$ is the time at which $\sigma_2 = \frac{1}{2}$.

The quality of the fit during the early recovery phase is shown in Fig. 23. In fitting the curve to each record the scale of the tension change was fixed by the step amplitude and the measured T_2 , and the only parameter left available for adjustment was $t_{\frac{1}{2}}$. In all the releases the fitted points fall on the records within the uncertainties due to noise, oscillations, etc., but in the stretch the fitted points are detectably above the record from about 0.5 to 2 ms and below the record at times greater than 4 ms.

No simple procedure of this kind is applicable in the period before the step is complete because the rate constants of the recovery processes must be expected to vary as the shortening progresses. In a later section (pp. 485 *et seq.*) the procedure is extended so that with the help of a computer it can be used during the period before completion of the step.

Reality of the fastest term. The fourth term in eqn. (5) is very poorly determined by the curves in Fig. 22 because it has decayed to a small fraction of its initial amplitude by the time of the earliest of the

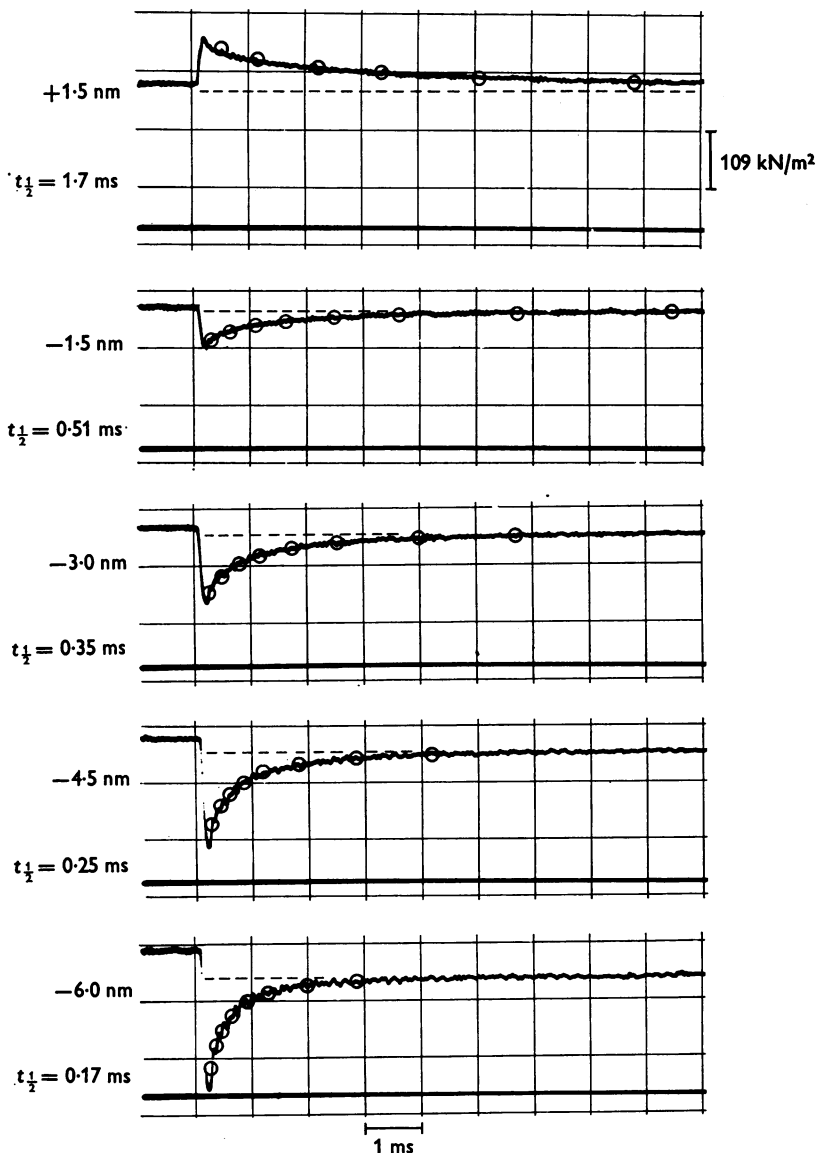


Fig. 23. Fitting a family of recovery records by a single 'standard' curve with adjustable time scale. Records similar to those in Fig. 12 but on a faster time base. Circles: values of tension T given by solving equation (4) with σ_s given by eqn. (5) with the value of $t_{1/2}$ stated at the left of the panel. Positions of asymptotes (T_s) shown by broken lines; they are horizontal except in the top panel. Expt. of 17 Jan. 1974; sarcomere length $2.2 \mu\text{m}$; temperature $+1.0^\circ\text{C}$.

measurements on which these curves are based (0.15 ms after the middle of the step). The reality of its existence becomes very evident when the shortening step is immediately followed by an equal lengthening step: the slower terms due to the shortening step have not had time to decay much by the time the lengthening step is applied and are therefore still nearly large enough to neutralize the corresponding components of the latter, whereas the fastest term in response to the stretch stands out because the corresponding component of the response to the shortening step has largely decayed away. Experiments of this kind will be presented in a later paper of this series.

A term with roughly this speed is also needed when the fitting procedure is extended to cover the period during the step itself (pp. 485–488). The amplitude of this term found necessary to match the initial tension change (p. 487) is substantially larger than would correspond to the value given in eqn. (5). The agreement of the fitted points to the records of shortening steps in Fig. 23 would not be appreciably altered by taking this larger amplitude for the fastest term; the only noticeable change would be that the earliest of the circles on the trace for the 1.5 nm stretch would be raised by about one third of its diameter, making even more evident the failure of this procedure to match the response to stretch.

Relation between half-time and step size. The fact that responses to shortening steps can be fitted by scaling the time axis provides a convenient way of comparing the rates of recovery in the responses to steps of different sizes. The standard curve was fitted to the double logarithmic plot of σ_2 from an experimental record as described on page 477 and the value of t corresponding to the 0.5 point on the standard curve was then taken as the half-time of the early recovery in the experimental record. Its reciprocal is a convenient measure of the rate of the early recovery process.

The results from six experiments are plotted against step amplitude in Fig. 24. The figure also shows a curve which corresponds to that fitted by Huxley & Simmons (1971*b*) to the results then available.

It is seen that the general trend is similar and the actual values in stretches are about the same, but in releases the rates appear substantially higher in the new results. The difference is largely or entirely due to the non-exponential form of the decay together with the fact that in the present experiments the half-time was measured earlier in the decay, both because of the sharper step (so that more of the early part of the response was accessible to measurement) and because the procedure measured the decay from T'_1 , not T_1 .

Phase 1 of the transient (initial tension change)

Fig. 14 showed that the initial tension change occurs more or less synchronously with the recorded length change of the part of the fibre between the markers. The fact that the extreme tension is reached slightly before the length change is complete is evidently due chiefly to the fact that the early tension recovery, which is the main feature of phase 2, has already begun. The lag in onset of the recorded tension change relative

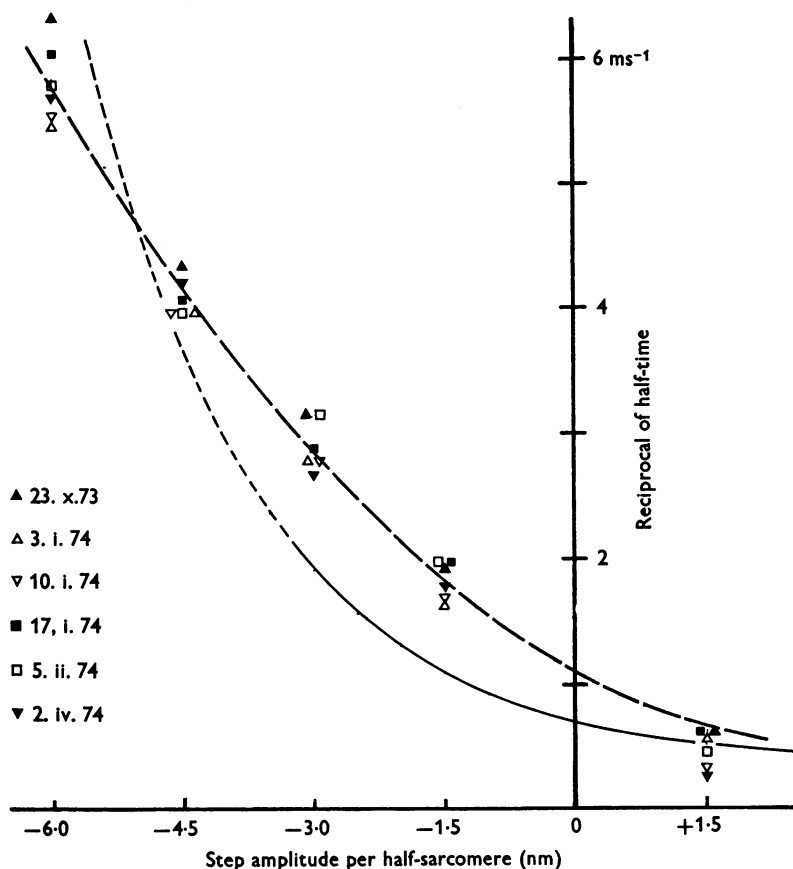


Fig. 24. Reciprocal of half-time of early recovery phase, plotted against amplitude of step. Points: estimated as described on page 479 from six experiments at about $+1^{\circ}\text{C}$. Dashed line: parabola fitted to the points from shortening steps, and used in the computer simulation of the early part of the transient (p. 486). Continuous line: 1.75 times the value of 'rate constant' fitted by Huxley & Simmons (1971b) to data from steps with 1 ms rise time. The factor 1.75 is the ratio of the times to drop to $1/e$ and to $\frac{1}{2}$ in the 'standard curve' (eqn. 5). Interrupted line: extrapolation of the same function to larger releases.

to the length change, and the slight hump visible half-way through the initial tension change in some of the records (e.g. the 3 nm release), are to be expected from the characteristics of the force transducer (compare Fig. 10). To a first approximation at least, the initial tension change therefore behaves as if the fibre contains an undamped compliance.

The T_1 curve

A simple way of displaying the characteristics of this compliance is to plot the extreme tension (T_1) against the amplitude of the step (y_s). This was done in Fig. 13 (steeper curve). The line is slightly concave upwards but approaches the horizontal axis at a finite angle, not tangentially or asymptotically. The steepness of the line has clearly been reduced by the onset of the early tension recovery. Further, the fact that this recovery is faster the greater the size of a shortening step (p. 474) must have introduced some upward curvature. The extent of these effects will be examined in a later section (p. 488).

The line is substantially steeper and less curved than what we reported on the basis of earlier experiments (Huxley & Simmons, 1971*a, b*). The difference is almost entirely due to the improvement in speed of the step length change (Fig. 19; p. 489) but a small part is attributable to the higher temperature (4 °C) in the experiments of 1970–1 (see next section).

Dependence of T_1 on isometric tension. Just as in the case of the T_2 curve (p. 473), it was found that there was little or no change in the curve of T_1/T_0 against step amplitude during the course of an experiment when runs under similar conditions were compared, in spite of considerable changes in T_0 . The value of T_1 for a given size of step thus varied in direct proportion to the isometric tension.

The line did, however, become substantially less steep when the temperature was raised and the isometric tension increased as a result. This effect is at least partly due to the increased rapidity of the early tension recovery, but is probably not entirely accounted for in this way (see pages 493 *et seq.*).

Tension change during the step itself

More precise information about the characteristics of the rapid elasticity should in principle be obtainable by examining the relation between tension change and length change during the early part of the step itself, when the early tension recovery will not have had time to affect the results greatly. To do this it was necessary to correct the recorded tension signal, as described on page 461, for (a) the response of the force transducer (Appendix F), (b) the force needed to accelerate the fibre mass (Appendix D) and (c) the force exerted on the fibre by the surrounding fluid.

(Appendix E). The amounts of these corrections in a typical case are shown in Fig. 25 (note different scales). The largest value of correction (*a*) is about 12% of the extreme tension change during the step; for correction (*b*) the corresponding figure is about 2% for and correction (*c*) about 0.5%.

The change of length at each instant could not be taken direct from the spot-follower signal because the latter was too noisy. The time course of the length change was assumed similar to that of the movement of the tip

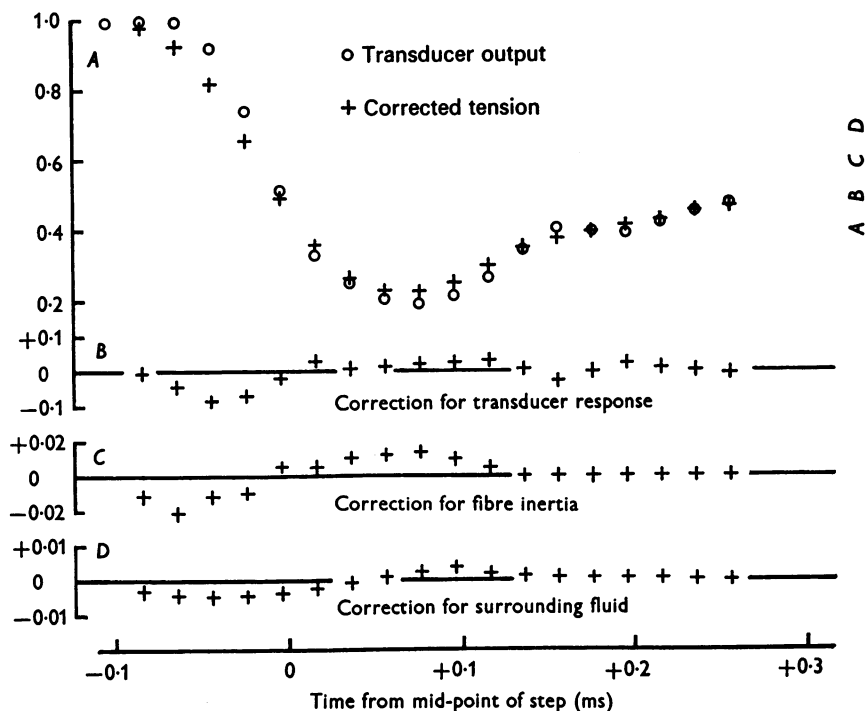


Fig. 25. Correction of recorded tension for response of force transducer, fibre inertia, and effect of surrounding fluid. *A*, circles, transducer output read at intervals of $20\ \mu\text{s}$; crosses, tension corrected for all three sources of error. *B*, correction for transducer response, on same scale. *C*, correction for fibre inertia, on 5 times larger scale than *A*. *D*, correction for effect of surrounding fluid, on 10 times larger scale than *A*. All expressed as fractions of value during isometric tetanus. 4.5 nm release in series used in constructing Fig. 26.

of the motor arm recorded in separate experiments with the servo in motor control (Methods, p. 454 and Fig. 5*B*); this is justifiable because at these short times the effective feed-back utilized by the servo is from the differentiated p_M signal and the condenser coupling from p_M to the spot-follower signal (Methods, p. 453). The exact time, relative to the timing

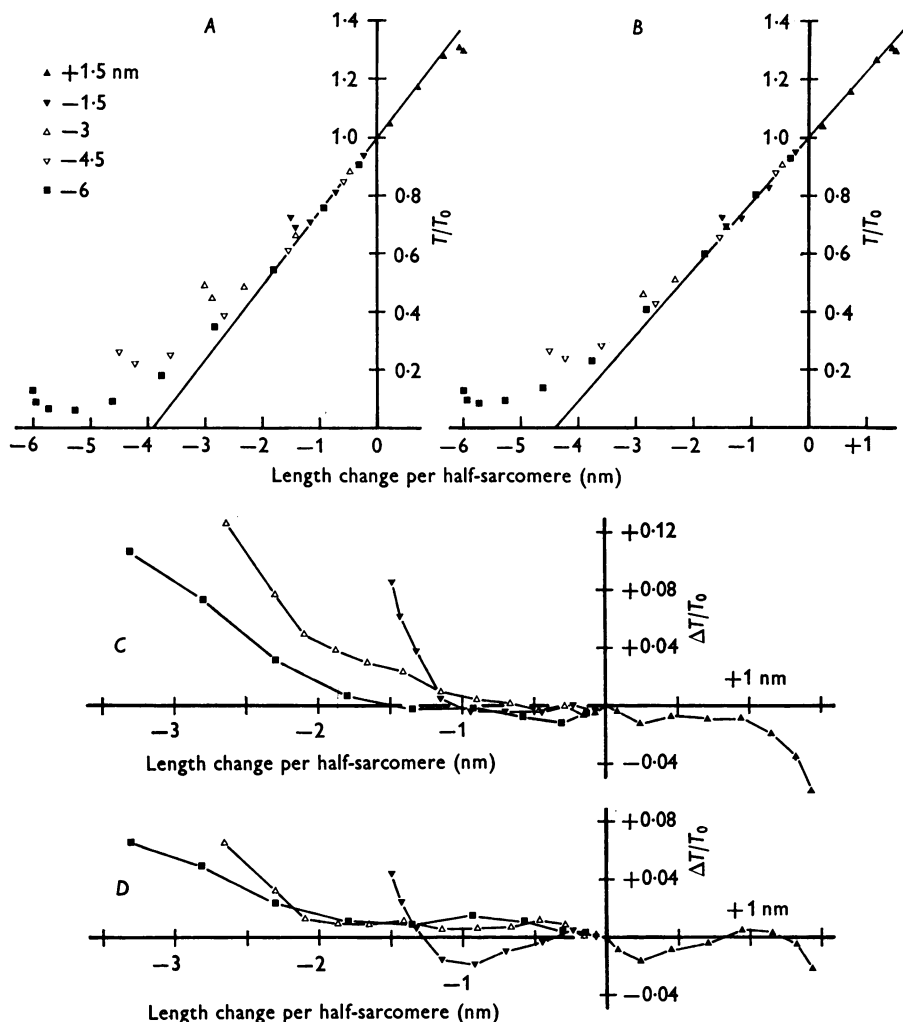


Fig. 26. Instantaneous tension plotted against length change during the step itself. Each symbol corresponds to a different step amplitude, as indicated. Experiment of 15 Nov. 1973; sarcomere length $2.2 \mu\text{m}$; temperature $+1.0^\circ\text{C}$. *A*, recorded signal from force transducer, corrected for transducer response, fibre inertia and fluid movement; straight line drawn by eye, intersecting horizontal axis at -3.9 nm . *B*, same points but in addition the tension change expected to result from the resting viscosity has been subtracted; straight line drawn by eye, intersecting horizontal axis at -4.4 nm . *C*, as *A*, but ordinate is vertical deviation from the straight line in *A*; larger scale, larger number of points but -4.5 nm step has been omitted for clarity. *D*, as *B*, but ordinate is vertical deviation from the straight line in *B*. In *A* and *C* the points from the early parts of steps of all amplitudes fall close to the line. In *B* and *D*, where the 'viscosity correction' has been applied, the points from steps of different amplitude no longer superpose well, and there are inflexions near the point corresponding to zero length change.

mark, at which the length change was 50% complete was judged in each experiment from a record of p_M taken during a step.

Fig. 26*A* shows the resulting plots of tension change against length change for one of the two experiments in which families of transients were recorded on a sufficiently fast time base (0.2 ms/cm). A closely similar result was obtained in the other experiment. It is seen that (i) the points from steps of different amplitude superpose well until the length change is about half complete (see also Fig. 26*C*), after which the early tension recovery causes the points to deviate upwards, as is to be expected, and (ii) there is no discontinuity or inflexion between the points corresponding to small amounts of stretch and of release. These features suggest that the rapid elasticity is, to the degree of precision reached in these experiments, a pure elasticity, without viscosity or static friction.

Absence of viscosity in stimulated fibres. Fig. 18 showed the tension change that would be produced by a step change of length acting on the passive properties of a resting fibre. The small component that persists after completion of the step decays with time constants that are much longer than the 100 μ s (half the rise time of the step) with which we are concerned here, and would therefore appear as a contribution to the stiffness; its value would be only about 0.3% of that represented by the slope of the points in Fig. 26*A*.

The large component, however, is proportional to rate of change of length, and if a correction is made for it, the shape of the curve, not only its steepness, will be changed. Fig. 26*B* and *D* show the effect of applying this correction. It is seen that the points from steps of different amplitude no longer superpose, and that an inflexion has been introduced at the origin. The tension change in the stimulated fibre therefore appears not to contain a component corresponding to this resting viscosity.

Events when tension reaches zero. When a shortening step of the usual time course, and amplitude of 7.5 nm or more per half-sarcomere, was applied, the output of the force transducer fell below the base line, as shown in Fig. 27. The result of applying the correction for transducer response is also shown: the fibre pushes on the transducer with a force of the order of 5% of the isometric tension for a period of 0.1 ms or more, according to the size of the step. From the moment when tension reaches zero, the distance between the markers (upper trace) shortens much more slowly than would correspond to the motion of the motor arm; presumably the clip at the motor end of the fibre is no longer in contact with the hook on the motor arm, and the whole of the force needed to decelerate the fibre, and also the tendon and clip at the motor end, appears at the force transducer, where the clip must have remained adherent to the hook. The time integral of the negative force on the transducer is about equal

to the decrease of momentum of the fibre, together with the tendon and clip at the motor end, estimated from the speed of the motor arm at the moment when tension drops to zero and the speed of shortening of the fibre (from the spot-follower signal) at the moment when the tension passes from a negative to a positive value. A precise comparison is not possible because of uncertainties, for example, in the mass of the tendon and clip, and in the elastic properties of the tendon when the tension is near zero. The correction procedures for the fibre inertia and fluid movement described in Appendices D and E are not applicable when the clip at the motor end is detached from its hook.

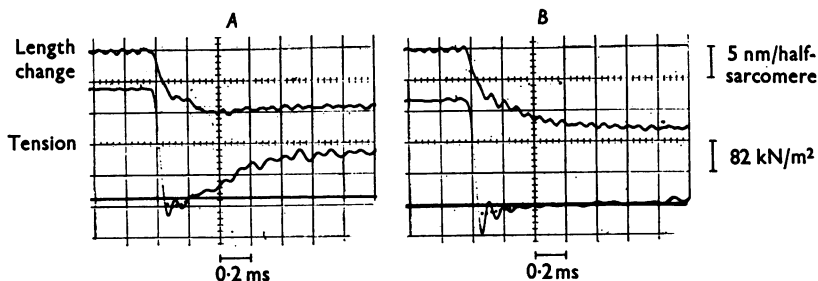


Fig. 27. Shortening steps in which tension goes below zero. Standard (0.2 ms) step applied under length control. Amplitude of step: 9 nm/half-sarcomere (A) and 12 nm/half-sarcomere (B). Spot-follower signal (upper trace) shows that fibre length ceases to follow step time course after tension reaches zero; the later part of the shortening occurs much more slowly. Dots: tension corrected for transducer response. Correction carried out with 10 μ s interval; each dot is mean of three consecutive values. Experiment of 15 Nov. 1973, sarcomere length 2.2 μ m; temperature +1.0 $^{\circ}$ C.

The time integral of the negative tension is several times larger than the estimated loss of momentum by the fibre itself. The whole length of the fibre must therefore be subject, during this period, to a negative tension nearly as great as that which appears at the force transducer. In spite of this, it shortens at a finite speed, at most about 12 μ m/ms in each half-sarcomere.

These observations imply that the structures responsible for the rapid elasticity do not become slack as the tension in them approaches zero (see Discussion, p. 503).

Computer simulation of phases 1 and 2

The procedure used on page 477 shows that the phase of early tension recovery after completion of a shortening step is adequately fitted by assuming a linear elasticity with a recovery curve (given by eqn. (5)) which is similar for all amplitudes of step but whose speed increases with step size. To test whether the same assumptions give a satisfactory fit during the step itself, it is necessary to take account of the change in the

rate constants of the recovery process that would be expected to occur as the shortening progresses. This requires the numerical solution of a set of differential equations, which we performed on the IBM 360/65 computer at University College London, as follows.

For the case of a rectangular step (which was tacitly assumed in the procedure used on page 477 for fitting the early recovery phase) the assumed behaviour would be given exactly by a system in which:

(1) the change of tension from its isometric value is the sum of four terms, $T_\alpha - T_\beta$, each of which obeys Hooke's law in response to an instantaneous length change. Their stiffnesses (for 1 half-sarcomere) are denoted by $A_\alpha T_0/y_0$, $A_\beta T_0/y_0$, etc., where y_0 is the instantaneous decrease of length needed to bring the total tension to zero from the isometric value T_0 , and $A_\alpha + A_\beta + A_\gamma + A_\delta = 1$;

(2) after completion of a step these terms decline exponentially towards values $A_\alpha(T_2 - T_0)$, etc., with rate constants $a_\alpha/t_{\frac{1}{2}}$, etc., equal to those in equation (5), $t_{\frac{1}{2}}$ having the value appropriate to the size of the step.

These postulates can be stated in the form

$$T = T_0 + T_\alpha + T_\beta + T_\gamma + T_\delta, \quad (6)$$

$$dT_\alpha/dt = (A_\alpha T_0/y_0) \cdot dy/dt - (a_\alpha/t_{\frac{1}{2}})\{T_\alpha - A_\alpha(T_2 - T_0)\}, \quad (7)$$

and equations similar to (7) for T_β , T_γ and T_δ .

To match the experimental results for different sizes of step, it is further necessary to assume that:

(3) T_2 varies with step amplitude y_s according to an equation such as

$$T_2/T_0 = 1 + 0.01167 (y_s/\text{nm}) - 0.003333 (y_s/\text{nm})^2, \quad (8)$$

which is an adequate fit to the T_2 curve in Fig. 13 for values of y_s up to -6 nm, the largest value before tension reaches zero during the step, and

(4) $t_{\frac{1}{2}}$ varies with step amplitude according to an equation such as

$$1/t_{\frac{1}{2}} = (1.101 - 0.390 (y_s/\text{nm}) + 0.0648 (y_s/\text{nm})^2) \text{ ms}^{-1}, \quad (9)$$

which is the equation of the dashed line in Fig. 24.

In order to extend the procedure of page 477 so as to give the tension at all times during a step of known time course as well as during the subsequent early tension recovery, we have assumed that eqn. (7) holds at all times, with T_2 and $t_{\frac{1}{2}}$ given by equations identical to (8) and (9) except that y_s , the step amplitude, is replaced by y ($= (s - s_0)/2$), the instantaneous value of the length change per half-sarcomere, which of course becomes equal to y_s when the step is complete.

The time course of the length change was assumed similar to the movement of the tip of the motor arm (Fig. 5B), which is complete at $t = 0.12$ ms (99% at 0.095 ms). The values taken for the A 's were:

$$A_\alpha = 0.152, \quad A_\beta = 0.335, \quad A_\gamma = 0.140, \quad A_\delta = 0.373.$$

The first three were chosen so that the sizes of the three terms at any value of t after completion of a 3 nm shortening step were proportional to the values of the first three terms in eqn. (5) at the same value of t , thus ensuring that the later parts of the responses are well matched. The fourth term is larger than is implied by eqn. (5) (see pp. 477–479); its value was chosen

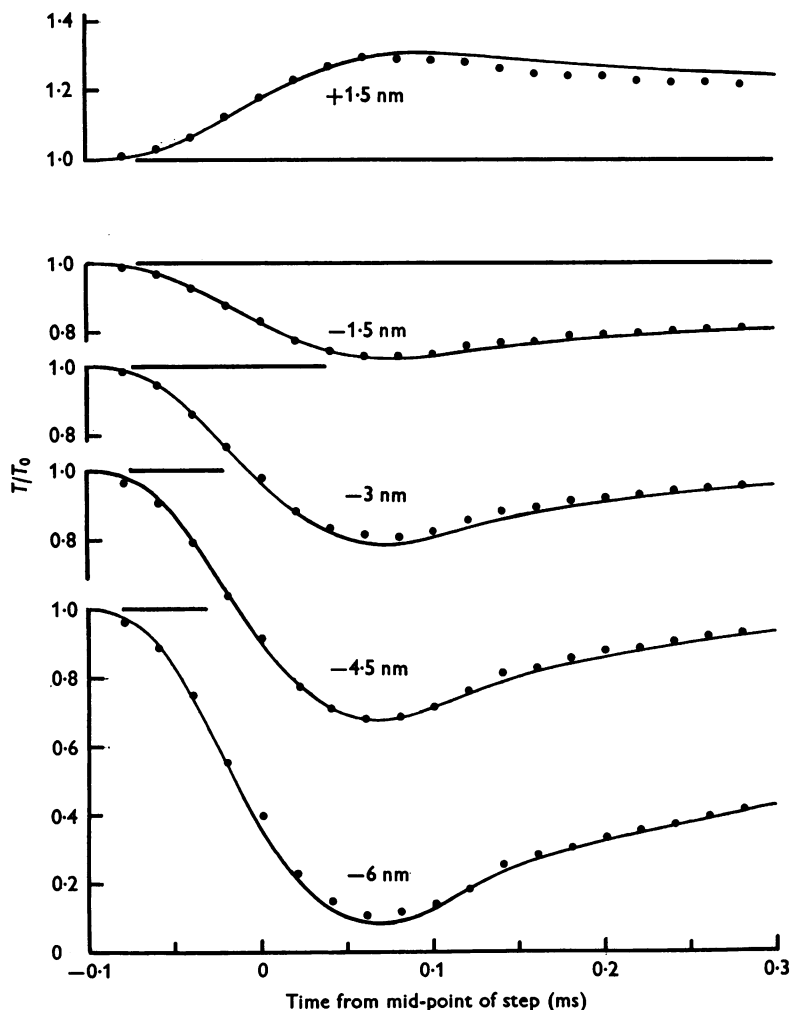


Fig. 28. Fitting the tension changes during a step by computer simulation. Lines: solutions of eqns. (6)–(9), with values given in the text (p. 486), for the indicated amplitudes of step per half-sarcomere. Dots: recorded tension in experiment of 2 April 1974, corrected for inertia and damping of transducer, inertia of the fibre itself, and inertia and viscosity of the Ringer solution. Sarcomere length $2.0 \mu\text{m}$; temperature $+2.5^\circ\text{C}$.

to obtain a match to the recorded tensions between about $t = 0$ and $t = 0.15$ ms.

The full lines in Fig. 28 show the time courses of tension computed on these assumptions when y_0 is taken as 4.0 nm. The points are corrected values of tension recorded in steps of corresponding amplitude in the experiment of 2 April 1974. There is good agreement in the shortening steps but, as is to be expected from the crude analysis on page 477 in the stretch the computed tension does not fall fast enough from its peak.

An equally good fit to the family of records of 15 Nov. 1973 (from which Fig. 26 is made) was obtained by taking $y_0 = 3.6$ nm and replacing eqn. (9) by

$$1/t_{\frac{1}{2}} = \{0.933 - 0.408 (y/\text{nm}) + 0.0907 (y/\text{nm})^2\} \text{ms}^{-1}, \quad (10)$$

which fits the half-times measured by the procedure of page 479 in this experiment.

We wish to emphasize that the only adjustments to parameters that were made in achieving this fit to the tension changes during the step itself were (a) the increase in A_s to a value (same for all experiments fitted) larger than corresponds to that in eqn. (5), and (b) the value for y_0 was adjusted for each experiment. All the other parameters (the a 's of eqn. (5); $t_{\frac{1}{2}}$ and T_2/T_0 as functions of y) were obtained from measurements on the tension records at times *after* completion of the step as described on pages 471–479.

Linearity of the instantaneous elasticity. The fitting procedure used in the previous section assumed that the instantaneous elasticity obeys Hooke's law. A fit nearly as good as that shown in Fig. 28 was obtained when a slight upward curvature was assumed in the instantaneous force-extension curve, according to the equation

$$T/T_0 = 1 + 0.244 (y/\text{nm}) + 0.003 (y/\text{nm})^2. \quad (11)$$

The values of the A 's had to be taken as $A_\alpha = 0.163$, $A_\beta = 0.359$, $A_\gamma = 0.150$, $A_s = 0.328$, i.e. the amplitude of the fastest-decaying term was reduced. The only noticeable defect of the fit to the shortening steps was that the extreme tension reached in the 6 nm step was not low enough, i.e. the T_1 curve was more curved than that found experimentally. It is doubtful whether the disagreement is outside experimental error.

A substantially larger deviation in the same direction appeared when the coefficient of $(y/\text{nm})^2$ was raised to 0.005 and the other parameters readjusted. This was probably beyond experimental error. No doubt a match could be obtained by, for example, allowing the rate constants of the four terms $T_\alpha - T_\beta$ to be different functions of length change, but we have not thought it worth while to go through such a procedure. Even if such an approach were adopted, the instantaneous length-tension

relation could not be more curved than the common asymptote to the sets of points in plots such as Fig. 26*A*. For the experiment analysed in Fig. 28, the most curved relation that could reasonably be assumed on this basis is

$$T/T_0 = 1 + 0.235 (y/\text{nm}) + 0.007 (y/\text{nm})^2, \quad (12)$$

which reaches zero at $y = -5.0$ nm, as against the value of -4.0 nm in the linear relation assumed in Fig. 28.

Computed response to slow steps. Fig. 19 showed the values of T_1 and T_2 in an experiment where the step was deliberately slowed so as to have approximately the same time course as those used in the experiments reported in 1971 (Huxley & Simmons, 1971*a, b*). The records are not on a fast enough time base to make possible a direct comparison with the expected time course as in Fig. 28, but we did compute the tension time courses by the same procedure, and the extreme tensions are plotted as the steeper lines in Fig. 19 for comparison with the observed T_1 's.

To match the responses of this fibre, the values taken for the A 's and a 's were identical to those used for Fig. 28 but

(a) the value for y_0 was 4.12 nm/half-sarcomere, chosen to fit the T_1 values in the fast (standard) steps;

(b) values for $t_{\frac{1}{2}}$ were taken as $0.9 \times$ those given by eqn. (9). This factor was needed in order to match the later part of phase 2 in the fast steps by the procedure described on pages 477 and 479;

(c) T_2 was assumed to be given by

$$T_2/T_0 = 1 - 0.0079167 (y/\text{nm}) - 0.0125 (y/\text{nm})^2 - 0.0005208 (y/\text{nm})^3. \quad (13)$$

This relation, which is shown as the full line among the T_2 points in Fig. 19, gives an adequate fit for shortening steps of amplitude up to 10 nm/half-sarcomere, the largest value used.

The T_1 values that were obtained when the length change was assumed to have the standard time course (complete in ≈ 0.2 ms) are plotted as the steepest curve in Fig. 19; it is seen to pass satisfactorily among the experimental points for fast steps.

The computation was repeated using the same parameters but with the step time course taken from the records of s during the slow steps. The resulting T_1 curve is shown as the middle line in Fig. 19. It is seen that it gives a good fit to the observed values for steps less than 6 nm in amplitude. The existence of a discrepancy in the larger steps is not surprising since the values of $t_{\frac{1}{2}}$ for length change beyond -6 nm were obtained by extrapolating the parabola fitted (Fig. 24) to values measured in steps not exceeding -6 nm in amplitude; the time course of phase 2 could not be measured in larger rapid steps because the tension dropped to zero. The

discrepancy suggests that the extrapolation of the curve of $1/t_1$ against y should be steeper than is implied by the fitted parabola.

Response to ramp shortening

The sharp initial drop of tension and the early recovery phase that are seen in response to stepwise shortening of a tetanized fibre are greatly smoothed out when the step is replaced by steady shortening of sudden onset (ramp shortening), and are represented only by the downward

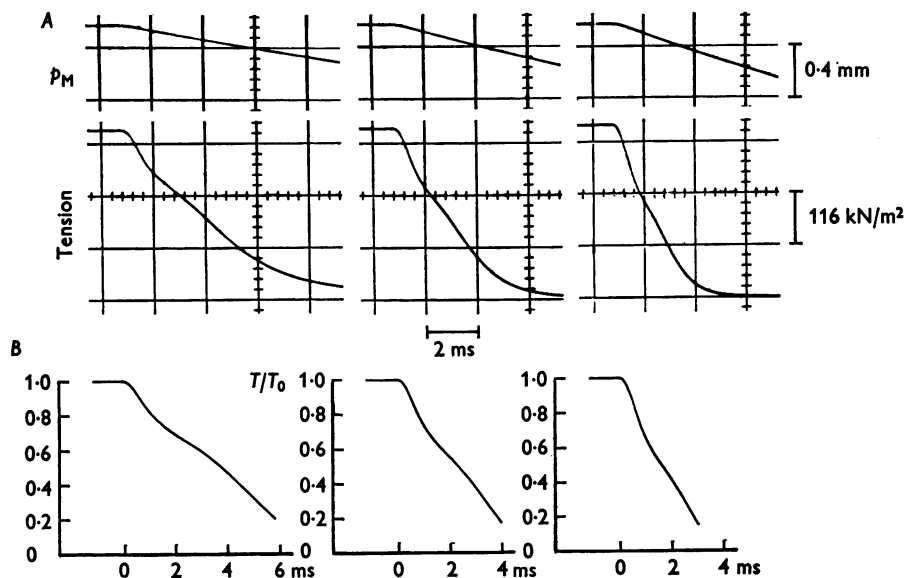


Fig. 29. Tension responses to ramp shortening at various speeds. *A*, in experiment of 24 May 1966. Upper traces, displacement of motor arm; lower traces, tension (force transducer critically damped with natural frequency 1.3 kHz approx.). Spot-follower and tendon clips not in use; steady shortening speed imposed by motor corresponds to 2.9, 4.2 and 5.6 nm/ms per half-sarcomere in the three panels from left to right, but actual shortening speeds within the fibre must have been appreciably lower on account of tendon compliance. Fibre from frog semitendinosus muscle, length 12 mm, striation spacing $2.1 \mu\text{m}$; temperature $+3.5^\circ\text{C}$.

B, computed, with time course of onset of shortening similar to that in *A*. Steady shortening speeds 2.0, 3.0 and 4.0 nm/ms per half-sarcomere. Rate constants as in eqn. (5) with $1/t_1$ given by eqn. (9); $A_x - A_d$ as on page 486; T_2 given by eqn. (13); $y_0 = 4 \text{ nm}$.

bulge in the early part of the responses reproduced in Fig. 29*A*. We did no experiments of this kind in this (1973-4) series; the records in Fig. 29*A* are from an experiment of 1966 by A.F.H. in collaboration with Dr C. M. Armstrong. We have not attempted to match the records in detail,

but the curves in Fig. 29*B* are computed, by the procedure described on pages 485–486, for conditions roughly comparable to those under which the corresponding records of Fig. 29*A* were obtained, and it is seen that the main features of the experimental records are satisfactorily reproduced.

For comparison with published records such as those of Bressler & Clinch (1974), we have also plotted a family of computed ramp responses against the instantaneous length change (Fig. 30*B*) instead of against time. The time course of onset of the ramp was made similar to that of Bressler & Clinch (1974, fig. 2). It is seen that the features visible in Fig. 29 are still evident. This way of plotting brings out the fact that as the shortening speed is increased the point of inflexion comes at a lower tension; this feature is also seen in the experimental records in Fig. 29*A*.

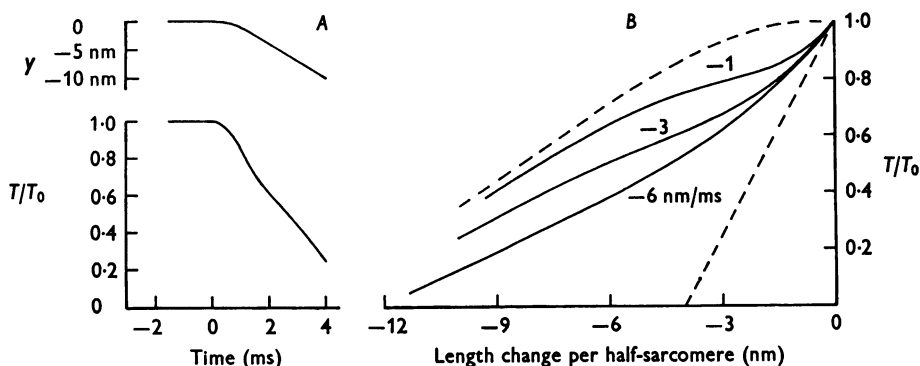


Fig. 30. Computed responses to ramp shortening with time course similar to that in records of Bressler & Clinch (1974, fig. 2); other parameters as in Fig. 29*B*. *A*, time course of shortening (upper curve) and tension (lower curve). Steady rate of shortening per half-sarcomere, 3 nm/ms. *B*, tension plotted against instantaneous value of length change. Full lines: tension computed for three values of steady shortening speed per half-sarcomere, 1, 3 and 6 nm/ms as indicated. Upper broken line: T_2/T_0 . Lower broken line: tension in response to instantaneous shortening ($y_0 = 4$ nm). The 3 nm/ms curve is from the same points as the curves in *A*.

As would be expected intuitively, all three of the computed force-extension curves of Fig. 30*B* lie between the T_2 curve (upper broken line) and the line representing the instantaneous stiffness (lower broken line). Each computed curve starts out from the isometric point with a slope nearly equal to that of the instantaneous stiffness, and deviates towards the T_2 curve as shortening progresses; the lower the shortening speed, the greater the time required to reach a given length change and therefore the greater the extent of this upward deviation.

The computed curves in Fig. 30*B* are approaching zero tension with

12–15 nm of shortening per half-sarcomere. In a muscle at a length equal to A. V. Hill's l_0 (sarcomere length probably 2.3–2.4 μm), these values would correspond to about 1.0–1.3 % of length.

Temperature effects in Phases 1 and 2

General description. Fig. 31 shows the early phase of tension transients at approximately 0, +4 and +8 °C. As is well known, the tetanic tension rises substantially over this range. However, the amplitude of the tension

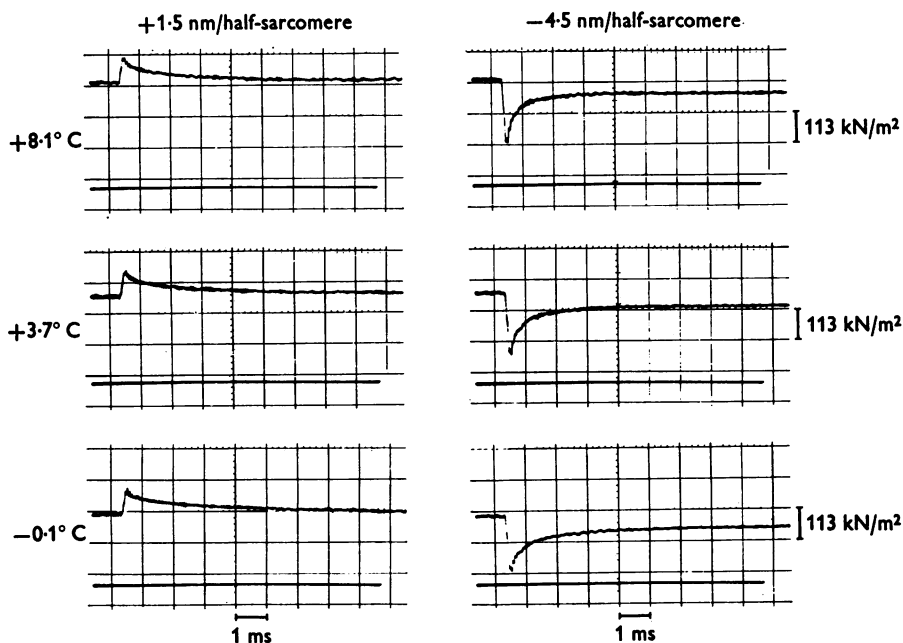


Fig. 31. Tension transients in response to lengthening and shortening steps at three temperatures. Lower line: base line for tension. Standard (0.2 ms) steps in length control. Experiment of 23 Oct. 1973. Sarcomere length, 2.2 μm .

response to a given step change of length is hardly altered, so that the curve of T_1/T_0 against step amplitude becomes less steep (Fig. 32). This figure also shows that there is a shift in the T_2 curve; considered as a change of position at given step size it is much less conspicuous than the change in T_1 but considered as horizontal displacements at any given fraction of the isometric tension the effects on T_1 and T_2 are roughly equal. Similar results were also obtained in one other experiment in the final series and in two experiments in July 1973, when the performance of the equipment was only slightly less good than that finally achieved.

It can be seen in Fig. 31 that the rate of recovery of tension during phase 2 increases with temperature. This is shown more clearly in Fig. 33 where plots of the early recovery phases at various temperatures are superposed. Because of this effect, the truncation of the tension change by the onset of recovery becomes more severe as the temperature is raised, thus

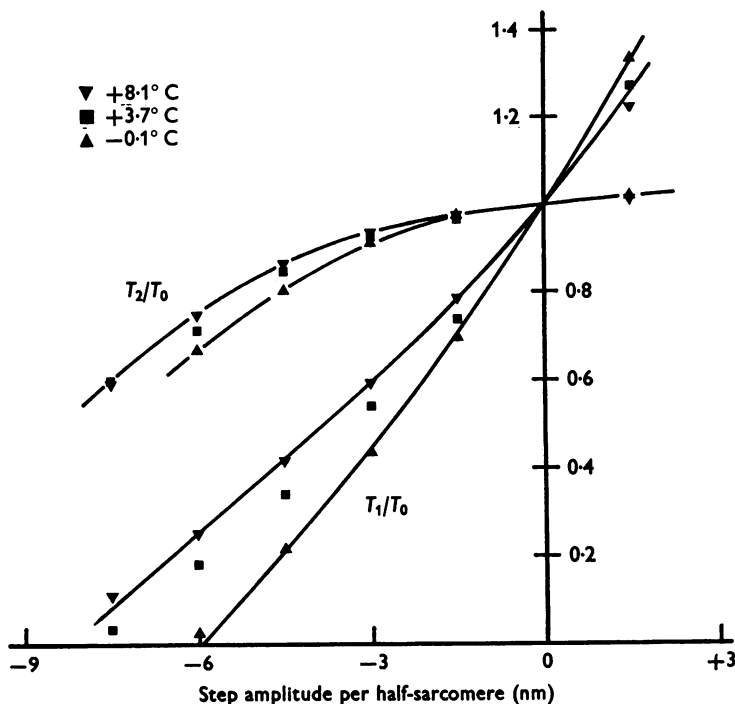


Fig. 32. Curves of T_1/T_0 and T_2/T_0 at different temperatures. From the same experiment as Fig. 31. The three runs were made in the order -0.1°C , $+3.7^\circ\text{C}$, $+8.1^\circ\text{C}$, and were preceded by a run at $+1.5^\circ\text{C}$ and followed by one at 1.1°C . The T_1/T_0 points from the two last-mentioned runs fell in the correct order between the corresponding points at -0.1 and $+3.7^\circ\text{C}$; the T_2/T_0 values were very close to each other and also to the values from the run at $+3.7^\circ\text{C}$. Lines through T_2/T_0 points drawn by eye. Lines through T_1/T_0 points: values computed for -0.1°C and $+8.1^\circ\text{C}$ by procedure described on p. 496, using Q_{10} s derived from phase 2 (p. 494) and assuming linear instantaneous elasticity.

reducing the slope of the T_1 curve. To test whether the whole of the apparent decrease in relative stiffness can reasonably be accounted for in this way, the quantity σ_2 (p. 475) was calculated for releases at different temperatures and plotted against time on log-log scales; an example is given in Fig. 34. In evaluating σ_2 (eqn. (4)), the value for T'_1 in the denominator was calculated by assuming that the slope of T'_1/T_0 against

step amplitude is independent of temperature and has the value obtained from T_1 in a 1.5 nm stretch at the lowest temperature used in the same experiment (-0.1°C in Fig. 34). It follows that if (a) the time course of the early recovery remained similar but on a faster time scale at the higher temperature, and if (b) the stiffness to an instantaneous step change of length were directly proportional to the increased tetanic tension, then it should be possible to match the curves by simply shifting the standard

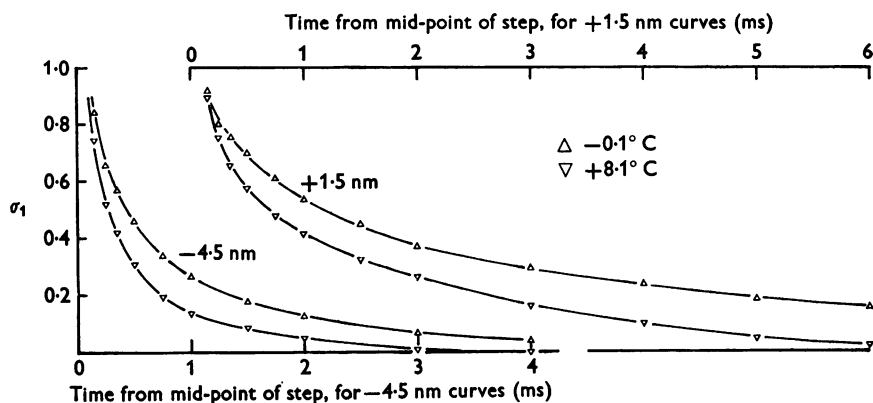


Fig. 33. Speed of early recovery phase at different temperatures. Ordinate: σ_1 (p. 474) from the corresponding records in Fig. 31. It is seen that, for each step amplitude, the recovery becomes more rapid as the temperature is raised.

curve along the time axis. It was found, however, that in most cases a satisfactory fit could not be obtained without also displacing the standard curve towards reduced amplitudes for the higher-temperature plots (Fig. 34). A fit judged to be within experimental error was obtained in all cases by shifts corresponding to (a) a value of 0.8 for q_1 , the Q_{10} of the ratio (instantaneous stiffness)/(isometric tension) and (b) a value of 1.85 for q_2 , the Q_{10} of the speed of the early recovery phase. These two measurements interact strongly with each other: if q_1 is taken as 0.7, the value found for q_2 becomes about 1.5, but the fit is not as good as with the values first mentioned. There is a suggestion that q_2 may be larger for small releases than for large ones.

Even stronger evidence that the stiffness does not increase in proportion to the rise of tetanic tension as the temperature is raised is provided by the responses to stretch. In the left-hand part of Fig. 31, the rise of tension in response to 1.5 nm stretch per half-sarcomere is almost identical in the top ($+8.1^\circ\text{C}$) and the bottom (-0.1°C) records, although the isometric tension is 50% greater in the former than in the latter. It is clear

from inspection of the traces that allowance for greater truncation at the higher temperature could not bring up the stiffness by anything like as much as 50 %. A modification of the procedure shown in Fig. 34 was applied to log-log plots obtained from these stretch responses and showed

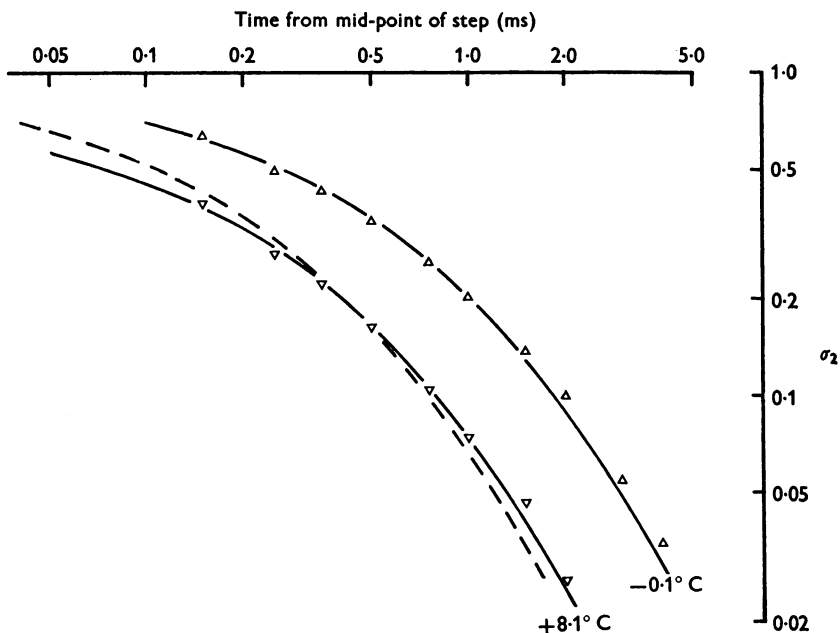


Fig. 34. Effect of temperature on amplitude and time-scale of early recovery phase of the tension transient. The quantity σ_2 (p. 475) calculated from the records in Fig. 31 of 4.5 nm releases are replotted on double logarithmic scales against time measured from the mid-point of the step. The right-hand continuous line is the 'standard curve', shifted horizontally (but not vertically) to give the best fit to the -0.1°C points; a satisfactory fit is achieved. The broken curve is the best fit that can be obtained to the $+8.1^\circ\text{C}$ points without a vertical shift; there are significant deviations at the two ends of the curve. The left-hand full line is the standard curve displaced downwards by a distance corresponding to a Q_{10} of 0.8 for the ratio of instantaneous stiffness to isometric tension, and then displaced horizontally for best fit; a satisfactory fit is again achieved.

that the effect of the allowance is 10–15 %. Values of q_1 and q_2 from stretches are uncertain because of the small number of experiments and the variability of the rate of the early recovery (p. 474); q_1 appears to be between 0.7 and 0.8, as found from the releases, while q_2 is probably higher than in the releases (2.0–2.5).

Computer simulation of temperature effects. We did not take records at raised temperature on a fast enough time-base to make possible a

comparison between corrected tension and solutions of eqn. (7) throughout the early part of each step, such as is shown in Fig. 28. We did, however compute the solutions, and the extreme tensions reached in steps of different size at -0.1°C and $+8.1^{\circ}\text{C}$ are shown as the steeper curves in Fig. 32. The values chosen for the A s and a s were identical to those used for Fig. 28 (see page 486) but, in order to match the responses of this fibre:

(a) the value taken for y_0 was $3.84\text{ nm/half-sarcomere}$ at -0.1°C , and varied with a Q_{10} of 1.25, the reciprocal of the Q_{10} of (stiffness/tension) found on p. 494 by analysing the later time courses of tension;

(b) values for $t_{\frac{1}{2}}$ at -0.1°C were 0.9 times those assumed for Fig. 28, and increased with temperature with a Q_{10} of 1.85, the value found on page 494 by analysing the later time course;

(c) at each temperature T_2 was given by a quadratic function of y fitted to the corresponding points in Fig. 32.

It is seen that the curves pass very close to the observed points for the same temperatures in the shortening steps, showing that the Q_{10} s derived from the later part of phase 2 account satisfactorily for the reduced steepness of the T_1 curve at higher temperatures. The agreement in the stretches is not good but this is to be expected as the tension time course in stretches near the peak is not well fitted by the solutions of eqn. (7) (Fig. 28). The line for $+3.7^{\circ}\text{C}$ would run almost exactly mid-way between those for -0.1°C and $+8.1^{\circ}\text{C}$, but most of the observed points lie somewhat higher than this, suggesting that the temperature effect may be greater in the lower part of the range. There is not enough data, however, to establish this point.

To match the T_1 points at 8.1°C without assuming an increase in y_0 , it was found necessary to take the rate constants as 3.1 times their values at -0.1°C .

DISCUSSION

An attempt to account in detail for the phenomena described here will be made in a later paper in this series, on the basis of the theory outlined by Huxley & Simmons (1971b). The following discussion will be restricted to matters more closely related to the interpretation of the experimental observations.

Relation to earlier observations

As pointed out earlier (page 464 and Table 4), the four phases into which the tension response to a step length change can be divided correspond to phases that are recognizable in the length changes which occur when the tension is lowered in a stepwise manner. The latter type of transient was first described by Podolsky (1960) and Civan & Podolsky (1966). Earlier

experiments by A. V. Hill and others on the whole muscles of frogs did not have good enough time resolution to detect phase 2 or to distinguish between phases 3 and 4.

Phases 2 and 3. The delayed rise of tension after a stretch and the delayed fall after a small release (phase 3) are strongly reminiscent of the corresponding responses of asynchronous insect muscles (Boettiger & Furshpan, 1954; Boettiger, 1957; Jewell & Rüegg, 1966), which are the basis of the oscillatory activity of that type of muscle (Pringle, 1949, 1967).

This type of response in a vertebrate muscle was first reported by Armstrong *et al.* (1966), using living isolated fibres from the semiten-dinosus muscle of the frog. It has since been seen in glycerinated preparations from numerous vertebrate muscles (Rüegg, Steiger & Schädler, 1970; Steiger, 1971; Heintz, 1972; Heintz, Kuhn & Rüegg, 1974). All these experiments have also shown a response similar to our phase 2 (partial early recovery of tension after completion of a step length change).

'Controlled release' experiments. In experiments by Hill (1950, 1953), Jewell & Wilkie (1958) and Bressler & Clinch (1974), moderately rapid shortening ramps were applied to frog or toad muscles, and the tension at each instant was plotted against the simultaneous change of length. None of these authors described the inflexion that is to be expected (Fig. 30) on the basis of our tension transients and which was seen in the few ramp releases that we carried out (e.g. Fig. 29A). However, the points for controlled releases (filled circles) in Jewell & Wilkie's fig. 8* do show inflexions not unlike those in our Figs. 29 and 30, and it seems probable that the phenomenon was present in those experiments, which were performed on frog sartorius muscles.

No such inflexion is detectable in the records of Bressler & Clinch (1974). The reason for this discrepancy is not clear; the curve is very sensitive to the characteristics of the force transducer and also to the speed of the muscle response, which may have been different in their experiments (toad sartorius) from ours (frog tibialis anterior).

There is, however, fair agreement as regards the amount of shortening needed to bring tension to zero in such 'controlled releases'. This was about 1.5% of muscle length in the experiments of Jewell & Wilkie (1958) and of Bressler & Clinch (1974), and 1.0–1.3% estimated from the curves in our Fig. 30 (see page 492), the slightly higher value in the whole-muscle experiments being very likely due to the contribution from tendon

* There is a discrepancy in the labelling of the curves in this Figure. Dr Jewell has told us that in the Figure itself the letter G should be exchanged with the letter E, and C should be exchanged with F. The legend is correct, and the filled circles do represent controlled releases.

compliance, etc., which in our experiments was eliminated by the spot-follower.

We conclude that the force-extension curve in a 'controlled release' experiment is the resultant of the processes which give rise to our T_1 and T_2 curves (together with some tendon compliance), features such as the early tension recovery and the non-linearities which are so conspicuous in our experiments being to a large extent smoothed out by using ramps instead of step releases. For the same reason, those features are not likely to be very conspicuous in the kinds of movement that occur *in vivo*, and it may be that a two-component system with an undamped series elastic component will be an adequate representation of active muscle for purposes such as the interpretation of movements in the whole animal, although not for understanding the mechanism of contraction.

Relation to 'series elastic component'. The rapid elasticity that we observe bears a resemblance to the 'series elastic component' (SEC) of the two-component theory of muscle (A. V. Hill, 1938) in that both are virtually undamped elements in series with other elements whose length can only change at a limited rate. However, apart from the fact that the rapid elasticity we are now discussing is about 3 times stiffer than the SEC, there are two major differences. First, after a step shortening which brings the tension to say $0.6T_0$, the rapid elasticity is re-stretched almost back to T_0 in a few milliseconds (Fig. 12), whereas the SEC is defined as being re-stretched by a contractile component whose speed of shortening is given by the steady-state force-velocity curve, a process which is about 50 times slower. Secondly, an essential feature of the SEC is that its force-extension curve was assumed fixed from a few milliseconds after the first stimulus of a tetanus onwards, whereas the stiffness of our rapid elasticity increases roughly in proportion to the rising tetanic tension (Huxley & Simmons, 1971*a*). Results of this type of experiment with the present form of our apparatus will be presented in a later paper. A similar result has been found by Bressler & Clinch (1974) who used ramp releases on whole toad sartorius muscles.

Further, the direct proportionality between the stiffness of the rapid elasticity and isometric tension when tetani at different sarcomere lengths are compared (Huxley & Simmons, 1971*a*; next paper of this series) implies that most of the rapid elasticity is a property of the contractile elements themselves.

The rapid elasticity must therefore not be thought of as something in series with and separate from the actual contractile system.

Absolute value of stiffness of rapid elasticity. The amount of *instantaneous* shortening per half-sarcomere needed to bring tension to zero from its isometric level, denoted by y_0 , is about 4 nm in the experiments described

here. This is about 0.4 % of fibre length, and is about one-third of the value found in ramp shortening at moderately high speed (pp. 490–492; Fig. 30; p. 497).

The value of y_0 reported by Huxley & Simmons (1971*b*) was 8 nm, i.e. twice as great as we now find. The difference is accounted for by (a) the greater speed of the length changes used in the present experiments (Fig. 19), (b) the fact that tension change during the step itself has been estimated (p. 481), as opposed to merely recording the extreme tension change, which is truncated by the early tension recovery, and (c) the lower temperature (see page 492). Julian & Sollins (1975), from measurements of tension transients with steps of intermediate speed (complete in about 0.4 ms), and without using a spot-follower, deduced an intermediate value for y_0 , about 5.8 nm per half-sarcomere.

The only other experiments on frog muscle with sufficient time resolution to give values for the stiffness of the rapid elasticity are those of Truong (1974) and of Schoenberg, Wells & Podolsky (1974). In both cases the direct experimental result is the mechanical propagation velocity; Truong obtained a maximum of 130 m/s (25 °C) while Schoenberg *et al.* obtained 175 and 220 m/s in *sartorii* of two subspecies of *Rana pipiens* (6 °C). The low value found by Truong, which implies a low stiffness, is likely to be related to the small tension that seems to have been developed by his preparations (20 g wt. in an adult sartorius, Fig. 4*B*). Schoenberg *et al.* calculate from their velocities an upper limit of stiffness corresponding to $y_0 = 4.5$ nm. This calculation included an allowance for the variation of fibre lengths within the muscle; further allowance for components of the muscle that might not be accelerated during the applied step brought the figure up to 5.5–8.0 nm per half-sarcomere. These figures are probably as close as can reasonably be expected to our value of 4 nm in view of the difference of method and the uncertainties in the corrections. In the present experiments short fibres were deliberately chosen to minimize propagation effects, reducing the expected lags in the tension responses to less than 25 μ s. This is too short to be usefully measured in our corrected tension records (e.g. Fig. 25). Thus, our results cannot be compared directly with those of Truong or Schoenberg *et al.*

Interpretation of early tension recovery in terms of elastic and viscous components

Behaviour qualitatively similar to phases 1 and 2 of the transient – an initial tension change which recovers partially – is given by either of the arrangements of two elastic components and one viscous component shown in Fig. 35. If the components have linear characteristics, then both arrangements give identical responses, provided that the values of the components

are appropriately adjusted; it would therefore be impossible to say that one arrangement fits a set of experimental data better than the other. However, the T_2 curve, which is represented by the steady-state force-extension relation of the network, is very non-linear, and the following two arguments show that this non-linearity is more simply explained by the arrangement of Fig. 35*B* (Voigt element) than by that of Fig. 35*A* (Maxwell element):

(1) In the Maxwell arrangement the characteristic of the M_1 component is given by the T_2 curve. The tensions in the two elastic components add; the nearly straight T_1 curve therefore requires that the M_2 component

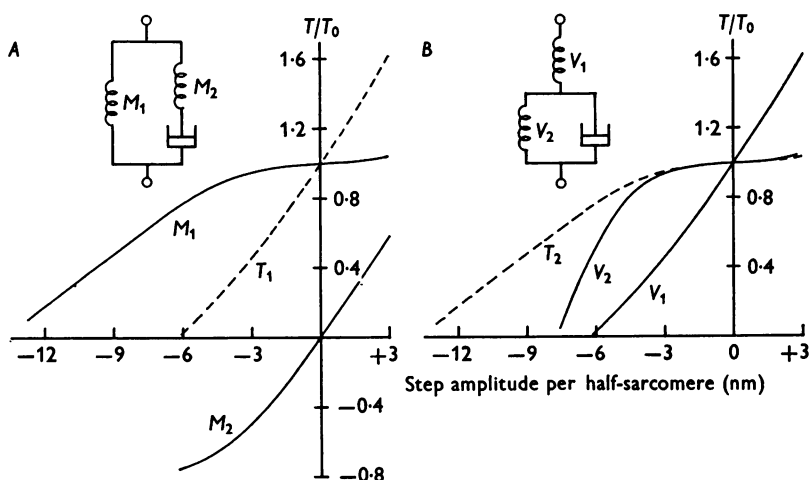


Fig. 35. Explanation of early tension recovery in terms of alternative arrangements of elastic and viscous components.

A, spring M_1 in parallel with series combination of dashpot and spring M_2 (Maxwell element). Full lines: force-extension curves of M_1 and M_2 ; that for M_1 is identical with the T_2 curve of Fig. 13. Interrupted line: T_1 curve of Fig. 13. At any value of the abscissa, the sum of the ordinates of the M_1 and M_2 curves equals the ordinate of the T_1 curve.

B, spring V_1 in series with parallel combination of dashpot and spring V_2 (Voigt element). Continuous lines: force-extension curves of V_1 and V_2 ; that for V_1 is identical with the T_1 curve of Fig. 13. Interrupted line: T_2 curve of Fig. 13. At any value of the ordinate, the sum of the abscissae of the V_1 and V_2 curves equals the abscissa of the T_2 curve.

has a non-linear characteristic whose curvature counteracts that of the M_1 component (see Fig. 35*A*). This seems an unlikely coincidence, especially as the characteristic of the true rapid elasticity appears to be very nearly straight (see page 488). No difficulty of this kind arises with the Voigt arrangement, since the T_1 curve is the characteristic of the

component V_1 alone; all that is needed is to assume that this has a nearly linear characteristic.

(2) When a second step is applied within 1 or 2 msec the stiffness of the fibre is hardly different from that in the first step (Ford *et al.* 1974). This is to be expected on the Voigt arrangement since in both cases the instantaneous stiffness is due to V_1 alone. With the Maxwell arrangement, however, the stiffness of M_2 is present in both cases but has added to it the stiffness of M_1 , proportional to the slope of the M_1 curve in Fig. 35*A*. This changes from nearly zero at $y = 0$, the working point for the first step, to about half the slope of the M_2 curve at $y = -6$ nm, the working point for a second step if the first has an amplitude of 6 nm per half-sarcomere. Hence in this case the stiffness in the second step should be about 50% greater than in the first. The experiments briefly reported by Ford *et al.* (1974) will be presented and discussed in detail in a later paper in this series.

These arguments certainly favour the Voigt rather than the Maxwell arrangement. Although they are probably not conclusive, we regard them as strong enough to justify us in continuing to discuss the system as if it consists of a rapid elasticity separable from and in series with an element which behaves like a very non-linear elasticity with damping. There are no grounds for thinking of the 'dashpot' in Fig. 35*B* as a component distinct from the 'spring' with which it is in parallel; the two together represent a structure whose length readjusts itself with a lag when the force on it is changed.

Characteristics of the rapid elasticity

Absence of viscosity. The evidence given on pages 484 and 488 suggests that the rapid elasticity responsible for phase 1 of the transient is nearly linear and nearly undamped. At the time we did the experiments, we did not expect to be able to analyse the tension records in this degree of detail; if we had, we would have taken fast records in each experiment and used still higher time-base speeds. As it is, the conclusion that the 'viscosity' of the active fibre is substantially less than the apparent viscosity of the resting fibre depends on a precision of timing of the tension change relative to the length change which is near the limit of what can be extracted from the records. The effect of a viscosity equal to that of the resting fibre would be to displace the steep part of the tension trace to the left by about 5 μ s. We think it unlikely that we would have failed to detect a shift of this amount, but it is not completely beyond the range of errors that could be imagined. The precision of the measurements is not enough to set a lower limit to the viscosity of an active fibre.

The absence of appreciable viscosity in stimulated fibres is consistent

with the relatively low damping of longitudinal waves propagating in whole muscles found by Truong (1974).

Another uncertainty in interpreting the apparently lower viscosity of active, as compared with resting, fibres is the point mentioned on page 469, that the 'viscosity' of a resting fibre may in fact be a stiffness which relaxes after step length change with an appreciable time constant. If this relaxation time were of the order of say $50\ \mu\text{s}$, then a 'viscosity' of this kind in a stimulated fibre, in parallel with the 'rapid elasticity', would appear as a small increase in the stiffness of that elasticity together with an increase in the amount of the fastest component of the early tension recovery. It would therefore not be recognised as a viscosity, and the absence of detectable viscosity in the responses of the active fibre would in that case not be evidence that a feature of the resting fibre had disappeared on stimulation. However, the evidence mentioned on p. 470 suggests that the resting 'viscosity' has a relaxation time so short that it would indeed appear as a viscosity if still present in a stimulated fibre, so its disappearance when the fibre is stimulated is probably a genuine phenomenon.

This disappearance on stimulation is reminiscent of the behaviour of the 'short-range elastic component' (SREC) of resting muscle discovered by D. K. Hill (1968). Hill was unable to detect a similar component in response to slow stretch or release of tetanically stimulated muscle, and L  nnergren (1971) showed that the stiffness of this component was greatly reduced by treatment with contracture-producing agents (raised potassium; caffeine) at concentrations below the thresholds for causing a rise of tension. It is therefore likely that the 'viscosity' of resting fibres and the SREC are two expressions of the same underlying mechanism. The speed of the ramps in our experiments was 10^3 – 10^4 times greater than was used by Hill and by L  nnergren when demonstrating the SREC, and the resulting tensions were an order of magnitude larger than the plateau reached by the SREC; it is therefore not to be expected that the SREC should have been detectable as such in our experiments.

We do not of course imagine that the rapid elasticity of stimulated muscle is completely undamped; if, as we suppose, most of the rapid elasticity is actually within the crossbridges, any change in its length must be accompanied by relative sliding movement of the thick and thin filaments which will be resisted by the viscosity of the liquid between them. A rough calculation of this resistance leads to a value for the viscous coefficient equal to a few hundredths of the observed 'viscosity' of resting fibres. This would be far below the limit of detection in our experiments on stimulated fibres, since its effect would be to displace the steep part of each tension record by a time of the order of $0.1\ \mu\text{s}$.

Linearity of the elasticity. It should in principle be possible to obtain the

whole of the shortening side of the force-extension curve from records such as are shown in Fig. 27 where the step is larger than is needed to bring the tension to zero. The procedure for correcting for the response of the force transducer, however, makes it impossible to obtain a corrected tension at a time closer to the moment when tension reaches zero than the analysis interval. Using $20\ \mu\text{s}$, this means that the last point is obtained when the tension is still large; if $10\ \mu\text{s}$ is used the errors on the individual tension readings are multiplied (see Appendix F) and the result is unreliable. A somewhat faster force transducer, and records on a faster time-base are needed; a good result should be attainable with the existing time course of the step. The limitation would then be uncertainty about the residual tendon compliance at low tensions, unless the spot-follower were sufficiently improved so that its output could be used directly as an instantaneous measure of length during the step itself. Meanwhile, the results presented on page 488 show that the degree of curvature of the force-extension relation cannot be great. If the coefficient of the quadratic term is 0.003 as in eqn. (11) the slope of the line is only 11 % less at zero tension than at the isometric tension; if the coefficient is 0.005 or 0.007 (eqn. (12)) the corresponding figures are 19 and 30 % respectively. It was shown that 0.005 is about the upper limit of possible values.

Behaviour of the rapid elasticity at zero tension. If the rapid elasticity were attributable to links which buckle freely under compression, the stiffness would fall to zero when there is zero tension in the fibre. If different links were to go slack at different degrees of shortening (as is to be expected on account of the non-integral relation between thick and thin filament periodicities), the over-all force-extension curve would be concave upwards, approaching the horizontal axis tangentially, as proposed by Blangé, Karemaker & Kramer (1972*b*). This feature is not shown by our ' T_1 curves' (Fig. 13); these are a first approximation to the force-extension curve of the rapid elasticity, and they approach the horizontal axis at a finite angle and not tangentially or asymptotically (page 481), suggesting that the elastic structures do not go slack. The same conclusion is suggested by the curves of corrected tension against length change during the step itself (Fig. 26*A*), although with the existing technique it was not possible to obtain corrected tensions at the stage of a large shortening step where tension was approaching zero. After the tension signal had become negative, the correction for transducer response could again be applied (Fig. 27), showing that an appreciable negative tension was being exerted on the transducer. It is impossible to evaluate the stiffness of the fibre exactly at this stage, since its length change was not known precisely and the unknown characteristics of the slack tendon would affect the result, but it seems clear that this amount of negative tension

could not have been obtained unless the fibre had a stiffness comparable to that which it showed while the tension was positive.

We conclude that all the structures responsible for the rapid elasticity can resist a negative tension, and do not become slack as tension approaches zero.

Importance of using short fibres

We chose the anterior tibial muscle (cf. Sten-Knudsen, 1953) partly because of the firm tendon attachments and the shape of its fibres, but principally because its fibres are only 5–6 mm long, about half the length of those of the semitendinosus muscle which we had used previously. As a result, the natural frequency of longitudinal oscillations was so high (about 16 kHz: Appendix B) that they were not detectably excited by the step. The corrections for fibre inertia (Appendix D) and for fluid motion during the step (Appendix E) are both proportional to the square of the number of sarcomeres in the fibre (N_{tot}), and would therefore have been four times larger if semitendinosus fibres had been used. These effects would probably have prevented us from carrying out the analysis described on pages 481–485 if we had continued to use semitendinosus fibres. It is unlikely, however, that we would have seen oscillations as conspicuous as those found by Blangé *et al.* (1972*a, b*) in whole muscles, both because the muscles they used are two or three times longer than frog semitendinosus fibres and because the damping effect of surrounding fluid (Appendix A and Appendix C) was not present in their experiments.

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APPENDIX A

Damping and extra inertia due to surrounding fluid and to the markers

Stokes (1851, p. 20) showed that a rigid plate oscillating in its own plane with frequency ν in a medium of density ρ_m , viscosity η , experiences a resistive force f per unit area which may be written in the form

$$f = \frac{\eta}{\lambda} \nu + \frac{\rho_m \lambda}{2} \frac{d\nu}{dt}, \quad (\text{A } 1)$$

where v is the instantaneous velocity and λ , the space constant for decline of amplitude of fluid movement with distance from the plate, is given by

$$\lambda = (\eta/\pi\rho_m\nu)^{\frac{1}{2}}. \quad (\text{A } 2)$$

The first term in (A 1) implies a viscous coefficient η/λ , and the second implies an increment $\rho_m\lambda/2$ in the effective inertia of the plate, both expressed per unit area.

For a frequency of 16 kHz (calculated in Appendix B for longitudinal oscillations in a stimulated fibre), with $\rho_m = 10^3 \text{ kg/m}^3$ and $\eta = 1.8 \times 10^{-3} \text{ N.s/m}^2$, equation (A 2) gives $\lambda = 6.0 \mu\text{m}$.

These results may be applied to the case of a muscle fibre oscillating parallel to its axis (the error due to neglecting the curvature is small: an exact treatment for a long cylindrical fibre, radius a , shows that the viscous coefficient is underestimated by a fraction roughly equal to $\lambda/2a$, which is 6% when $a = 50 \mu\text{m}$, while the error in the added mass is of a smaller order of magnitude).

If the shape of the fibre cross-section is such that its perimeter is θ times the square root of its cross-sectional area, the surface per sarcomere becomes $s\theta\sqrt{A}$ and the viscous coefficient per half-sarcomere, expressed per unit reference cross-sectional area, is

$$b_m = \frac{s\theta\sqrt{A}}{2A_r} \left(\frac{\eta}{\lambda} \right)$$

or

$$b_m = \frac{\theta\eta}{2\lambda} \left(\frac{s \cdot s_r}{A_r} \right)^{\frac{1}{2}} \quad (\text{A } 3)$$

or

$$b_m = \frac{\theta}{2} \left(\frac{s \cdot s_r}{A_r} \pi\rho_m\eta\nu \right)^{\frac{1}{2}}. \quad (\text{A } 4)$$

The added inertia per unit length is $\theta\sqrt{A} \cdot \rho_m\lambda/2$; dividing by A gives the increment in effective density of the fibre as

$$\begin{aligned} \Delta_1 \rho_t &= \theta\rho_m\lambda/2\sqrt{A} \\ &= \frac{\theta\rho_m\lambda}{2} \left(\frac{s}{A_r s_r} \right)^{\frac{1}{2}}. \end{aligned} \quad (\text{A } 5)$$

For a cylindrical fibre this reduces to $\rho_m\lambda/a$.

The effective density of the fibre is also increased by a contribution $\Delta_2\rho_t$ due to the markers. The mass of each marker together with the grease attaching it to the fibre and the Ringer solution moving with it was estimated as $6 \mu\text{g} + (0.3 \text{ mm}^2) \times \rho_m\lambda/2$; $\Delta_2\rho_t$ is obtained by taking twice this quantity and dividing by the fibre volume.

The quantity $\rho_t + \Delta_2\rho_t$, denoted by ρ'_t , ranged from 1190 kg/m^3 in the largest fibre ($A_r = 18900 \mu\text{m}^2$, $N_{\text{tot}} = 2670$, $\rho_t = 1060 \text{ kg/m}^3$, $\theta = 4$,

$s = 2.1 \mu\text{m}$, $\lambda = 6.0 \mu\text{m}$) to 1381 kg/m^3 in the smallest ($A_r = 8360 \mu\text{m}^2$, $N_{\text{tot}} = 2450$), and the corresponding values of $\rho_t'' = \rho_t' + \Delta_1 \rho_t$ were 1278 and 1512 kg/m^3 .

APPENDIX B

Frequency of longitudinal oscillations

A stimulated muscle fibre has a nearly instantaneous elasticity represented roughly by the steepness of the relation between T_1 and y_s (Fig. 13) and evaluated more exactly on pages 481–484. Together with the mass of the fibre, this will determine a velocity of propagation for longitudinal displacements of the fibre substance. One would therefore expect the fibre to be capable of longitudinal oscillations (cf. Blangé *et al.* 1972*a, b*) unless considerable damping is present.

The compliance per unit length is C_t/L_{tot} and the mass per unit length is M/L_{tot} , so the velocity is $\sqrt{[(L_{\text{tot}}/C_t) \times (L_{\text{tot}}/M)]} = L_{\text{tot}}/\sqrt{[MC_t]}$. The period of the lowest-frequency mode of oscillation ($1/\nu_t$) is the time for a wave to travel from one end of the fibre to the other and back, i.e. $2\sqrt{[MC_t]}$.

Now $M = \rho_t A_r s_r N_{\text{tot}}$ and $C_t = 2N_{\text{tot}} y_0 / P_0 A_r$, where y_0 is the amount of instantaneous shortening per half-sarcomere needed to bring tension to zero. The effect of the inertia of the markers and of the fluid which moves with the fibre is to replace ρ_t by ρ_t'' (Appendix A), so

$$\nu_t = \frac{1}{2N_{\text{tot}}} \sqrt{\frac{P_0}{2\rho_t'' s_r y_0}}. \quad (\text{B } 1)$$

Now $s_r = 2.1 \mu\text{m}$, $y_0 \approx 4 \text{ nm}$ (p. 488), $\rho_t'' \approx 1400 \text{ kg/m}^3$ (Appendix A) and typical values of N_{tot} and P_0 are 2700 and 250 kN/m^2 respectively, giving $\nu_t = 19.1 \text{ kHz}$. This value is an overestimate because tendon compliance will lower the frequency; it can be shown that if $c = 0.2$, the calculated frequency is reduced from 19.1 to 15.9 kHz. This frequency should be less in a fibre which has been stretched because the stiffness is less: P_0 declines in proportion to overlap (Gordon *et al.* 1966*b*) but y_0 stays nearly constant (Huxley & Simmons, 1971*a*; next paper of this series). In a fibre stretched to $3.0 \mu\text{m}$ per sarcomere, P_0 is reduced to rather less than half so the frequency should be reduced by about 30%.

APPENDIX C

Damping of longitudinal oscillations

Three of the phenomena mentioned in this paper can be expected to cause damping of longitudinal oscillations, namely (1) the 'viscosity' of the fibre, (2) the early tension recovery and (3) the viscous component of the force due to the surrounding fluid. When the ends of a fibre are

fixed, the first harmonic of a longitudinal oscillation is given by the following relation between the longitudinal displacement u and the distance from one end of the fibre expressed as a fraction α of the total length:

$$u = u_{\max} \sin \pi \alpha \exp \mu t, \quad (\text{C } 1)$$

where $\mu = -\zeta + 2\pi i \nu_t$, ν_t being the frequency of the oscillation and $1/\zeta$ the time constant with which its amplitude decays.

Hence
$$\frac{\partial^2 u}{\partial \alpha^2} = -\pi^2 u, \quad \frac{\partial u}{\partial t} = \mu u, \quad \frac{\partial^2 u}{\partial t^2} = \mu^2 u, \text{ etc.} \quad (\text{C } 2)$$

After completion of a step displacement, the equation of motion is

$$N_{\text{tot}} \left(\rho_1'' s_r \frac{\partial^2 u}{\partial t^2} + 2b_m \frac{\partial u}{\partial t} \right) = \frac{\partial P}{\partial \alpha}. \quad (\text{C } 3)$$

P may be taken as equal to $P_0 + P_e + P_v$ where P_0 is the mean (isometric) tension, P_e is the component of tension change due to the 'instantaneous stiffness' and the 'early tension recovery', and P_v is the component due to the fibre viscosity.

In the absence of tension recovery,

$$P_e = \frac{P_0}{2N_{\text{tot}} y_0} \frac{\partial u}{\partial \alpha}, \quad \text{whence} \quad \frac{\partial P_e}{\partial t} = \frac{P_0}{2N_{\text{tot}} y_0} \frac{\partial}{\partial t} \frac{\partial u}{\partial \alpha}.$$

The recovery may be introduced by adding $-\gamma P_e$ to the right-hand side, γ being the initial rate constant for tension recovery, yielding

$$\left(\frac{\partial}{\partial t} + \gamma \right) P_e = \frac{P_0}{2N_{\text{tot}} y_0} \frac{\partial}{\partial t} \frac{\partial u}{\partial \alpha}. \quad (\text{C } 4)$$

From the definition of β (p. 468)

$$P_v = \frac{\beta}{2N_{\text{tot}}} \frac{\partial}{\partial t} \frac{\partial u}{\partial \alpha}. \quad (\text{C } 5)$$

Combining eqns. (C2)–(C5) yields

$$\mu^2 + \left\{ \frac{\pi^2 \beta}{2N_{\text{tot}}^2 \rho_1'' s_r} + \gamma + \frac{2b_m}{\rho_1 s_r} \right\} \mu + \frac{\pi^2}{2N_{\text{tot}}^2 \rho_1'' s_r} \left\{ \frac{P_0}{y_0} + \beta \gamma \right\} + \frac{2b_m \gamma}{\rho_1'' s_r} = 0. \quad (\text{C } 6)$$

Solving this equation for μ and taking the imaginary part gives, if the damping terms are small,

$$\nu_t = \frac{1}{2N_{\text{tot}}} \sqrt{\frac{P_0}{2\rho_1'' s_r y_0}}, \quad (\text{C } 7)$$

in agreement with eqn. (B1), Appendix B.

The real part gives

$$\zeta = \frac{\pi^2 \beta}{4N_{\text{tot}}^2 \rho_1'' s_r} + \frac{\gamma}{2} + \frac{b_m}{\rho_1'' s_r}, \quad (\text{C } 8)$$

the three terms representing respectively the three sources of damping originally mentioned.

If $N_{\text{tot}} = 2700$, $\rho_f'' = 1400 \text{ kg/m}^3$ (Appendix A), and β is assumed to have the value $3 \times 10^8 \text{ N.s/m}^3$ found in resting fibres, the first term amounts to 35 ms^{-1} . The evidence on page 484 suggests, however, that the viscosity of a stimulated fibre is much less than this, but it does not provide a lower limit. Rough calculations of the viscous resistance to sliding lead to values of the order of a few hundredths of the resting viscosity, in which case the contribution to the rate constant of decay of longitudinal oscillations would be around $1\text{--}2 \text{ ms}^{-1}$.

To estimate the second term, γ may be taken as $(1/\sigma_2)/(\text{d}\sigma_2/\text{d}t)_{t=0}$ with σ_2 given by the 'standard curve' (page 477) but with the A s adjusted to the values used in the computer simulation. This gives $\gamma = 3.0/t_{\frac{1}{2}}$, and $1/t_{\frac{1}{2}} = 1\text{--}2 \text{ ms}^{-1}$ in small releases (Fig. 24), so the second term is about 2 ms^{-1} .

From eqn. (A 4) in Appendix A, the third term becomes

$$\frac{\theta}{2\rho_f''} \sqrt{\frac{s\eta\rho_m\eta\nu_f}{s_r A_r}}.$$

For the smallest fibre, with $A_r = 8360 \mu\text{m}^2$, $\rho_f'' = 1512 \text{ kg/m}^3$ (see Appendix A), $\theta = 4$, $s = s_r = 2.1 \mu\text{m}$, $\eta = 1.8 \times 10^{-3} \text{ N.s/m}^2$ and $\nu_f = 16 \text{ kHz}$, this amounts to 4.4 ms^{-1} , and for the largest fibre ($A_r = 18900 \mu\text{m}^2$, $\rho_f'' = 1278 \text{ kg/m}^3$) to 3.4 ms^{-1} .

APPENDIX D

Non-oscillatory effects of inertia in the fibre

Consider a uniform fibre which obeys Hooke's law, the compliance of its complete length being C_f . Let C_0 and C_1 be the compliances of its attachments at the force-transducer end and the motor end respectively, and $c = (C_0 + C_1)/C_f$. If the motor arm is given a steady displacement Δp , the resulting extension of the fibre itself is

$$\Delta L_{\text{tot}} = \Delta p \cdot C_f / (C_f + C_0 + C_1) = \Delta p / (1 + c) \quad (\text{D } 1)$$

and the extension of the transducer attachment is $\Delta L_{\text{tot}} \cdot C_0 / C_f$. (D 2)

The total displacement z_0 of an element (extent $\delta\alpha$) of the fibre whose distance from the transducer end is a fraction α of the total fibre length is therefore given by

$$z_0 = \left(\alpha + \frac{C_0}{C_f} \right) \Delta L_{\text{tot}}. \quad (\text{D } 3)$$

If Δp is not steady but is undergoing a constant acceleration, the displacement z will be different; denote the difference by $u(\alpha)$ so that

$$z = z_0 + u(\alpha) = (\alpha + C_0/C_f) \Delta L_{\text{tot}} + u(\alpha). \quad (\text{D } 4)$$

A solution with u independent of time exists since equation (D 6) below contains no term which is time-dependent. In this solution the force required to accelerate the element is

$$M\delta\alpha\ddot{z} = M\delta\alpha(\alpha + C_0/C_I)\ddot{L}_{\text{tot}}. \quad (\text{D } 5)$$

This force is provided by the difference in tension between the two ends of the element, $(1/C_I)(d^2u/d\alpha^2)\delta\alpha$, so that

$$\frac{d^2u}{d\alpha^2} = M\ddot{L}_{\text{tot}}(\alpha C_I + C_0). \quad (\text{D } 6)$$

Single and double integration from $\alpha = 0$ to $\alpha = 1$ provide two equations, and the conditions for continuity of tension at the two ends of the fibre provide two more, between which we can eliminate $u(1)$, $(du/d\alpha)_{\alpha=0}$ and $(du/d\alpha)_{\alpha=1}$ to yield

$$\frac{1}{C_0}u(0) = \frac{-M\ddot{L}_{\text{tot}}}{6} \left(1 + \frac{2C_I(C_0 + C_1) + 6C_0C_1}{C_I(C_0 + C_1 + C_I)} \right). \quad (\text{D } 7)$$

This is ΔT , the amount by which the force on the transducer differs from the tension that the fibre would exert on it if the motor displacement was the same but the acceleration was zero.

Writing $c = (C_0 + C_1)/C_I$ as before, and assuming $C_0 = C_1$, (D 7) becomes

$$\Delta T = \frac{-M\ddot{L}_{\text{tot}}}{6} \left(1 + 2c \frac{(4 + 3c)}{4 + 4c} \right) \quad (\text{D } 8)$$

$$\approx \frac{-M\ddot{L}_{\text{tot}}}{6} (1 + 2c). \quad (\text{D } 9)$$

Now $M = \rho_t A_r s_r N_{\text{tot}}$, or $\rho'_t A_r s_r N_{\text{tot}}$ if the effect of the markers is included (Appendix A), so the excess force per unit reference cross-sectional area is

$$\Delta P = \frac{\Delta T}{A_r} \approx -\frac{\rho'_t s_r N_{\text{tot}}}{6} (1 + 2c) \ddot{L}_{\text{tot}}. \quad (\text{D } 10)$$

The fraction c may be estimated from the gain needed during the experiment to match the spot-follower signal to the signal from the motor sensor; from the gains and scale values shown in Fig. 2 it follows that $c = 10^4 G_L / N_{\text{tot}} - 1$.

To a fair degree of approximation, the change of length during the step is proportional to the right-hand side of equation (1) (p. 454), and the total change of length in the step is $2N_{\text{tot}}y_s$.

Hence

$$\ddot{L}_{\text{tot}} = -(\pi/d)^2 N_{\text{tot}} y_s \sin \pi t/d \quad \left(-\frac{d}{2} < t < \frac{d}{2} \right) \quad (\text{D } 11)$$

and

$$\Delta P = + \left(\frac{\pi N_{\text{tot}}}{d} \right)^2 \frac{\rho'_t s_r y_s}{6} (1 + 2c) \sin \pi t/d. \quad (\text{D } 12)$$

This expression has its extreme values at the start and completion of the step, when $\sin \pi t/d = \mp 1$. Taking $\rho'_t = 1300 \text{ kg/m}^3$ (Appendix A), $N_{\text{tot}} = 2700$, $d = 0.2 \text{ ms}$, $y_s = -3 \text{ nm}$ and $c = 0.2$, these values are $\pm 3.4 \text{ kN/m}^2$, about 2.5% of the observed tension change in a release of 3 nm per half-sarcomere. The direction of the effect is to cause a delay in the onset of the recorded tension change and an overshoot at the peak.

The discontinuities implied by equation (D 12) at the start and completion of the step are of course not real, both because the steady state assumed in this treatment has not then had time to be established and because eqn. (1) is not an exact description of the step.

APPENDIX E

Force during step due to surrounding fluid

The force due to the surrounding fluid during and after an imposed 'step' displacement may be calculated as follows. Stokes (1851, p. 102) showed that if a plate moves in its own plane with velocity $v = v(t')$, an arbitrary function of time, then the force per unit area experienced by the plate at time t is

$$f = -(\rho_m \eta / \pi)^{\frac{1}{2}} \int_{-\infty}^t \frac{\dot{v}(t')}{\sqrt{[t-t']}} dt'. \quad (\text{E } 1)$$

Using eqn. (1) (p. 454) as an approximation for the time course of displacement during a step of amplitude p_s complete in time d , we have

$$v = v(t') = \frac{\pi p_s}{2d} \cos \frac{\pi t'}{d},$$

whence

$$f_1 = \frac{\pi^2 p_s}{2d^2} \left(\frac{\rho_m \eta}{\pi} \right)^{\frac{1}{2}} \int_{-\frac{1}{2}d}^t \frac{\sin \pi t'/d}{\sqrt{[t-t']}} dt' \quad (\text{E } 2)$$

for times during the step displacement ($-d/2 < t < d/2$), and

$$f_2 = \frac{\pi^2 p_s}{2d^2} \left(\frac{\rho_m \eta}{\pi} \right)^{\frac{1}{2}} \int_{-\frac{1}{2}d}^{+\frac{1}{2}d} \frac{\sin \pi t'/d}{\sqrt{[t-t']}} dt' \quad (\text{E } 3)$$

after the step is complete ($t > \frac{1}{2}d$).

Writing $U = \pi(t-t')/d$, eqn. (E 2) becomes

$$f_1 = \frac{\pi p_s}{2d} \left(\frac{\rho_m \eta}{d} \right)^{\frac{1}{2}} \left\{ \sin \frac{\pi t}{d} \int_0^{\pi(t/d+\frac{1}{2})} \frac{\cos U}{\sqrt{U}} dU - \cos \frac{\pi t}{d} \int_0^{\pi(t/d+\frac{1}{2})} \frac{\sin U}{\sqrt{U}} dU \right\} \quad (\text{E } 4)$$

and f_2 is equal to the same expression except that the lower limit of the integrals is $\pi(t/d - 1/2)$ instead of zero. The integrals are Fresnel integrals, tabulated for example by Jahnke & Emde (1943, p. 35) with the symbols

$$C(z) = \frac{1}{\sqrt{[2\pi]}} \int_0^z \frac{\cos U}{\sqrt{U}} dU, \quad S(z) = \frac{1}{\sqrt{[2\pi]}} \int_0^z \frac{\sin U}{\sqrt{U}} dU.$$

Hence

$$f_1 = \frac{\pi p_s}{d} \left(\frac{\pi \rho_m \eta}{2d} \right)^{\frac{1}{2}} \left\{ \sin \frac{\pi t}{d} \cdot C \left(\frac{\pi t}{d} + \frac{\pi}{2} \right) - \cos \frac{\pi t}{d} \cdot S \left(\frac{\pi t}{d} + \frac{\pi}{2} \right) \right\}. \quad (\text{E } 5)$$

Similarly,

$$f_2 = \frac{\pi p_s}{d} \left(\frac{\pi \rho_m \eta}{2d} \right)^{\frac{1}{2}} \left\{ \sin \frac{\pi t}{d} \left[C \left(\frac{\pi t}{d} + \frac{\pi}{2} \right) - C \left(\frac{\pi t}{d} - \frac{\pi}{2} \right) \right] - \cos \frac{\pi t}{d} \left[S \left(\frac{\pi t}{d} + \frac{\pi}{2} \right) - S \left(\frac{\pi t}{d} - \frac{\pi}{2} \right) \right] \right\}. \quad (\text{E } 6)$$

The quantities inside the curly brackets, denoted by ψ , are plotted against t/d in Fig. E 1. The extreme values of ψ are 0.6 in the resisting direction at $t = -0.21d$, when the displacement is 20% complete, and 0.53 in the direction of the movement at the moment when it is just complete.

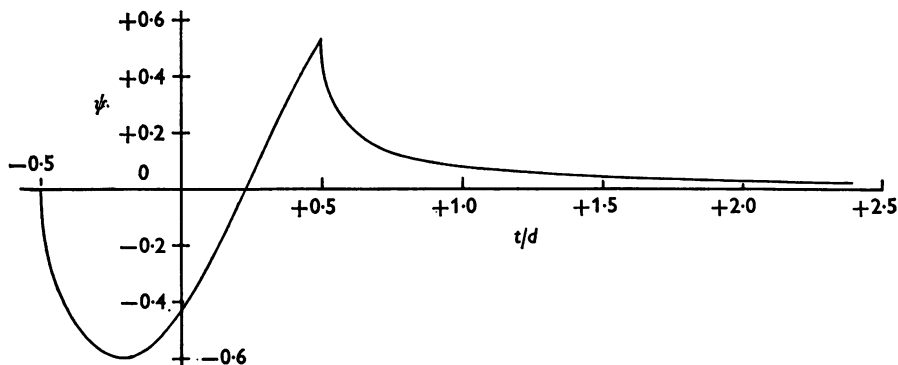


Fig. Appendix E 1. Time course of force due to surrounding fluid on unit area of fibre surface undergoing a 'step' displacement according to eqn. (1), complete in time d . For definition of ψ , see text.

By an argument parallel to that used for obtaining eqn. (D 9), the force appearing at the transducer will be equal to the total surface area of the fibre multiplied by the expression given in eqn. (E 5) or (E 6) with p_s replaced by $(1 + 2c)/6$ times the change in total length of the fibre, i.e. $(1 + 2c) N_{\text{tot}} y_s / 3$. The total area is $N_{\text{tot}} s \theta \sqrt{A}$ (cf. Appendix A), so the force per unit reference cross-sectional area is

$$\begin{aligned} \frac{N_{\text{tot}}}{A_r} s \theta \sqrt{A} \frac{\pi}{d} \frac{(1 + 2c) N_{\text{tot}} y_s}{3} \left(\frac{\pi \rho_m \eta}{2d} \right)^{\frac{1}{2}} \psi \\ = \frac{\theta N_{\text{tot}}^2}{3} \left(\frac{s \cdot s_r}{A_r} \frac{\pi^3 \rho_m \eta}{2d^3} \right)^{\frac{1}{2}} (1 + 2c) \psi \cdot y_s. \quad (\text{E } 7) \end{aligned}$$

For a step of 3 nm per half-sarcomere complete in $d = 0.2$ ms, with $c = 0.2$ as in Appendix D, and with the other values as in Appendix A, the coefficient of ψ comes to 1.14 kN/m² for the largest fibre and to 1.44 kN/m² for the smallest.

APPENDIX F

Correction of output of force transducer

If the effective stiffness, damping coefficient, and mass of the moving part of the transducer are k_t , b_t , and m_t , respectively, the displacement x of the moving part will be governed by the equation

$$k_t x + b_t \dot{x} + m_t \ddot{x} = T, \quad (\text{F } 1)$$

where T is the instantaneous value of the force exerted on the transducer by the fibre. The transducer output signal ξ varies linearly with x . If the force is steady,

$$T = K(\xi - \xi_0), \quad (\text{F } 2)$$

where ξ_0 is the signal when the force is zero and K is a calibration factor. From eqn. (F 1) this must be equal to $k_t x$, so

$$x = K(\xi - \xi_0)/k_t. \quad (\text{F } 3)$$

Differentiating and substituting into (F 1) gives, for the general case,

$$T = K(\xi - \xi_0 + \frac{b_t}{k_t} \dot{\xi} + \frac{m_t}{k_t} \ddot{\xi}) \quad (\text{F } 4)$$

$$= K(\xi_{\text{corr}} - \xi_0) \quad (\text{F } 5)$$

where

$$\xi_{\text{corr}} = \xi + \frac{b_t}{k_t} \dot{\xi} + \frac{m_t}{k_t} \ddot{\xi} \quad (\text{F } 6)$$

and K is the calibration factor obtained with a steady force.

The coefficients b_t/k_t and m_t/k_t are obtained as follows. When the transducer is unloaded, eqn. (F 1) becomes

$$m_t \ddot{x} + b_t \dot{x} + k_t x = 0 \quad (\text{F } 7)$$

whose solution is a damped sinusoidal oscillation of frequency

$$\nu_t = (1/2\pi) (k_t/m_t - b_t^2/4m_t^2)^{\frac{1}{2}} \quad (\text{F } 8)$$

decaying with time constant

$$\tau_t = 2m_t/b_t. \quad (\text{F } 9)$$

Combining (F 8) and (F 9) yields

$$m_t/k_t = 1/((2\pi\nu_t)^2 + (1/\tau_t)^2) \quad (\text{F } 10)$$

and

$$b_t/k_t = (2/\tau_t) \times (m_t/k_t). \quad (\text{F } 11)$$

With $\nu_t = 10.8$ kHz and $\tau_t = 0.10$ ms (Table 3) these become

$$m_t/k_t = 2.12 \times 10^{-10} \text{ s}^2 \quad (\text{F } 12)$$

and

$$b_t/k_t = 4.24 \times 10^{-6} \text{ s}. \quad (\text{F } 13)$$

The transducer output was read from the oscilloscope trace at time intervals h (usually $20 \mu\text{s}$, occasionally $10 \mu\text{s}$). Denoting three consecutive readings by ξ_- , ξ and ξ_+ , $\dot{\xi}$ is taken as $(\xi_+ - \xi_-)/2h$ and $\ddot{\xi}$ is taken as $(\xi_+ - 2\xi + \xi_-)/h^2$. With $h = 20 \mu\text{s}$, and using (F 12) and (F 13), (F 6) becomes

$$\xi_{\text{corr}} = 0.64\xi_+ - 0.06\xi + 0.42\xi_-. \quad (\text{F } 14)$$

With $h = 10 \mu\text{s}$, the corresponding result is

$$\xi_{\text{corr}} = 2.33\xi_+ - 3.24\xi + 1.91\xi_-. \quad (\text{F } 15)$$

If the standard deviation of a single value of ξ is σ , the standard deviation of ξ_{corr} is 0.77σ when $h = 20 \mu\text{s}$ and 4.4σ when $h = 10 \mu\text{s}$.

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